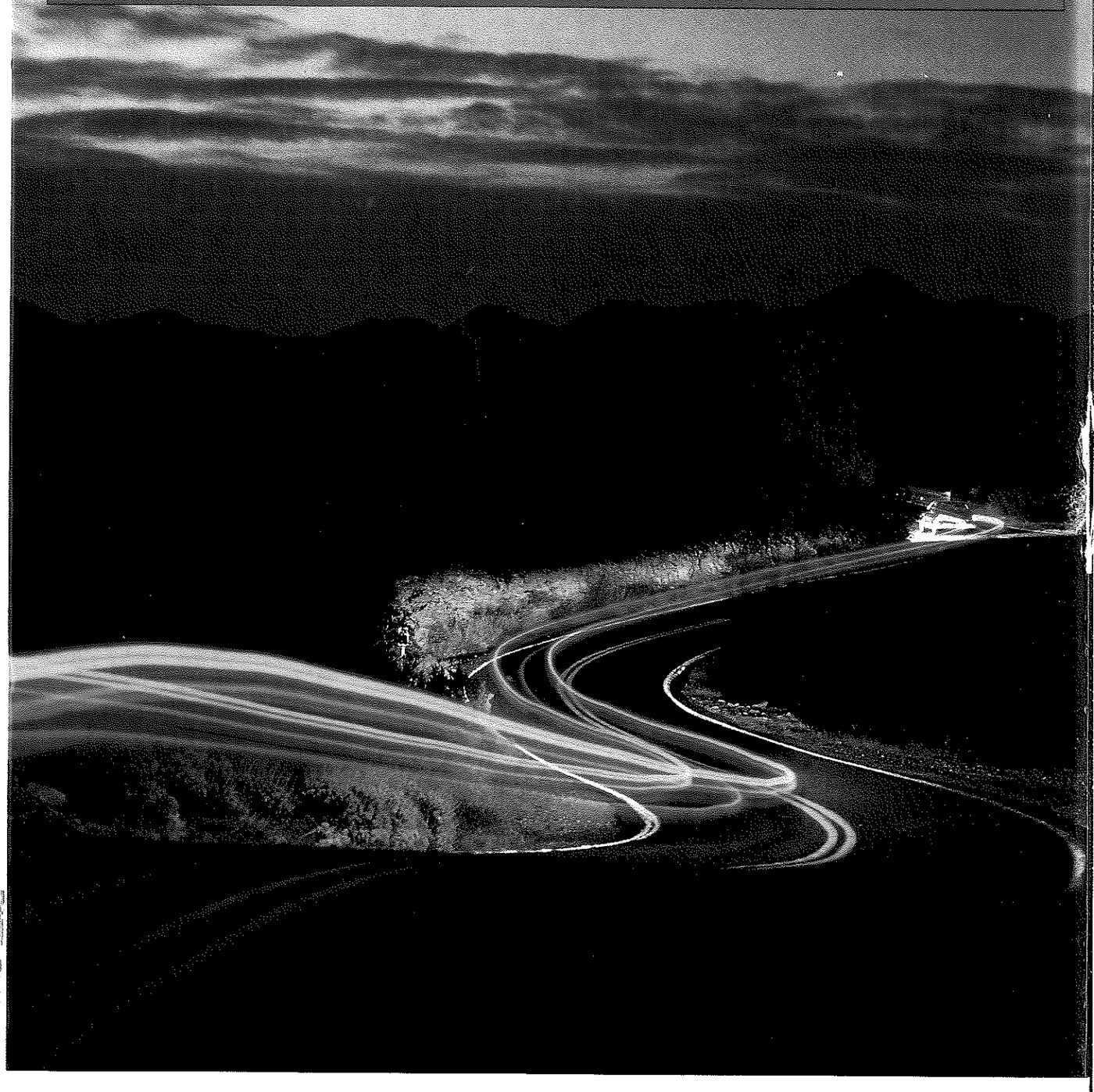
FOURTH EDITION

FUNDAMENTALS OF PHYSICS

EXTENDED



DAVID HALLIDAY • ROBERT RESNICK • JEARL WALKER

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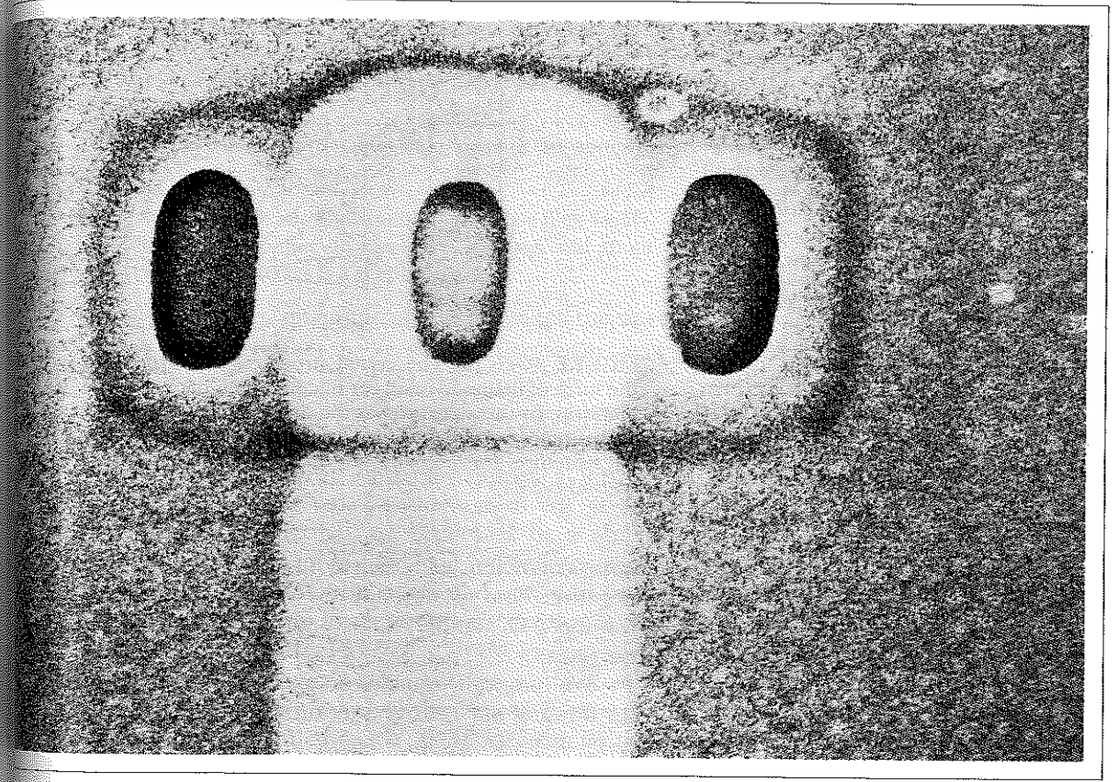
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CONDUCTION OF ELECTRICITY IN SOLIDS

46



This experimental transistor, shown in a scanning electron microscope image and made by scientists of the IBM Research Division, is the world's smallest transistor: the ovals you see are only 150 nm long. Today's memory chips have a capacity of 16 megabits. With this new transistor, memory chips may soon have a capacity of 4 gigabits or even more, which will dramatically increase the capabilities of computers of all sizes. But what is a transistor and how does it function?

46-1 THE PROPERTIES OF SOLIDS

We have seen how well quantum theory works when we apply it to individual atoms. In this chapter we hope to show, by a single broad example, that this powerful theory works just as well when we apply it to aggregates of atoms in the form of solids.

Every solid has an enormous range of properties that we can choose to examine. Is it transparent? Can you hammer it out into a flat sheet? What kinds of waves travel through it and at what speeds? Does it have interesting magnetic properties? Is it a good heat conductor? What is its crystal surface? Does it have special surface properties? . . . The list goes on. We choose here to focus on a single question: "What are the mechanisms by which a solid conducts, or does not conduct, electricity?" As we shall see, the laws that govern electrical conduction are quantum laws.

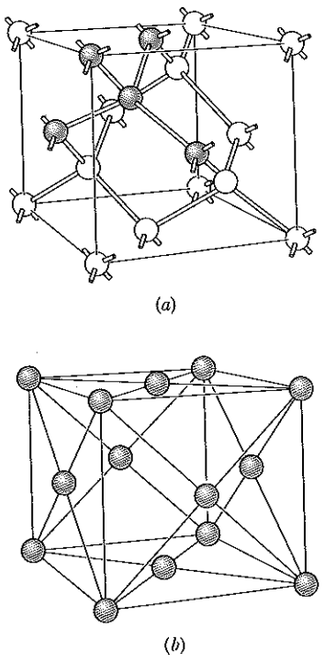
46-2 ELECTRICAL CONDUCTIVITY

In studying electrical conductivity, we choose to examine only solids whose atoms are arranged to form a periodic three-dimensional lattice. Figure 46-1 shows such lattice structures for carbon (in the form of diamond), silicon, and copper. We shall not consider such materials as plastic, glass, or rubber, whose atoms are not arranged in any such regular way.

The basic electrical measurement that we can make on a sample is its *electrical resistivity* ρ at room temperature; see Section 28-4. By measuring ρ at various temperatures, we can also obtain a value for α, the *temperature coefficient of resistivity*. Finally, by making Hall effect measurements (see Section 30-4) we can find a value for n, *the number of charge carriers per unit volume* in the material being tested.

From measurements of the room temperature resistivity alone, we quickly discover that there are some materials—we call them **insulators**—that for all practical purposes do not conduct electricity at all. Diamond, for example, has a resistivity of about $10^{16}\ \Omega \cdot m$, greater than that of copper by a factor of about 10^{24}.

We can use our measured values of ρ, α, and n to divide most noninsulators into two major categories: **metals,** such as copper, and **semiconductors,** such as silicon. As we see from Table 46-1, a typical semiconductor (silicon), compared with a typical conductor (copper), (1) has far fewer charge carriers, (2) has a considerably larger resistivity, and (3) has a temperature coefficient of resistivity that is both large and negative. That is, although the resistivity of a metal *increases* with temperature, that of a semiconductor *decreases.*



(*a*)

(*b*)

FIGURE 46-1 (*a*) The crystal structures for carbon (in the form of diamond) and for silicon, which happen to be identical. In this structure, as the darkened spheres show, each atom is bonded to four of its neighbors. (*b*) The crystal structure for copper, an arrangement called *face-centered cubic.*

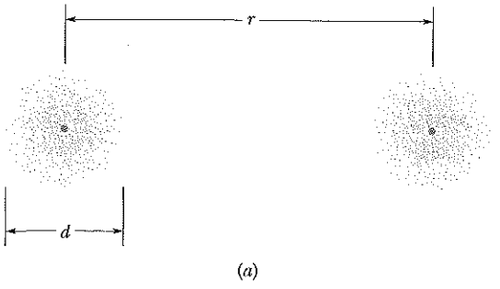
We have now established an experimental basis for framing our central question about the conduc-

TABLE 46-1
SOME ELECTRIC PROPERTIES OF TWO MATERIALS[a]

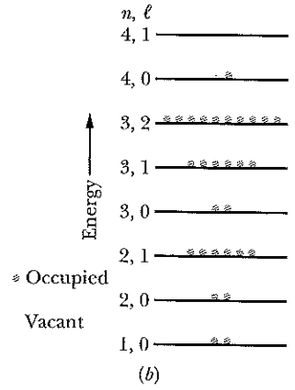
	UNIT	COPPER	SILICON
Type of conductor		Metal	Semiconductor
Density of charge carriers,[b] n	m^{-3}	9×10^{28}	1×10^{16}
Resistivity, ρ	$\Omega \cdot m$	2×10^{-8}	3×10^{3}
Temperature coefficient of resistivity, α	K^{-1}	$+4 \times 10^{-3}$	-70×10^{-3}

[a]All values are for room temperature.

[b]The value for the semiconductor includes both electrons and holes.



(a)



(b)

FIGURE 46-2 (*a*) Two isolated copper atoms, their electron clouds represented by dot plots. (*b*) Each atom has 29 electrons distributed over a set of energy levels as shown. The levels are identified by the notation n,ℓ, where n is the principal quantum number and ℓ the orbital quantum number. Each energy level contains $2(2\ell + 1)$ quantum states, defined by the quantum numbers m_ℓ and m_s. For simplicity, the levels are shown here as being uniformly spaced in energy.

tion of electricity in solids. We pose it in specific terms:

What is there about diamond that makes it an insulator, about copper that makes it a metal, and about silicon that makes it a semiconductor?

As we shall see, quantum physics provides the answers to this question.

46-3 ENERGY LEVELS IN A SOLID

The distance between adjacent copper atoms in solid copper is 260 pm. Consider, as in Fig. 46-2*a*, two copper atoms that are separated by a much greater distance than this. As Fig. 46-2*b* shows, each of these isolated atoms has associated with it an array of discrete quantum states, each state defined by its unique set of quantum numbers. In the ground state of the neutral copper atom, its 29 electrons occupy the 29 states of this array that are lowest in energy, each state containing but a single electron as the Pauli exclusion principle requires.

If we bring the atoms of Fig. 46-2*a* closer together, they will—speaking loosely—gradually begin to sense each other's presence. In the formal language of quantum physics, their wave functions will begin to overlap. This overlap will occur first for the wave functions of the valence electrons, which, because they spend most of their time in the outer regions of the electron cloud of the isolated atom, are the first to make contact.

When the wave functions overlap, we no longer speak of two independent and isolated systems but of a single two-atom system containing $2 \times 29 = 58$ electrons. The Pauli principle requires that each of these electrons must occupy a *different* quantum state. The only way that this can happen is for each energy level of the isolated atom to split into *two* levels for the two-atom system.

We can bring up further atoms and in this way gradually construct a lattice of solid copper. If our specimen contains N atoms, each level of the isolated copper atom must be split into N levels. In this way, each *level* of the isolated atom becomes a **band of levels** in the solid. In a typical solid, an energy band is a few electron-volts wide. Since N is of the order of the Avogadro number, we can see that the individual energy levels within a band are very close together indeed.

Figure 46-3 suggests the band structure of the levels in a hypothetical solid in which we have assumed, for simplicity, that the bands do not overlap. The gaps between the bands represent ranges of energy that no electron may possess. In much the same

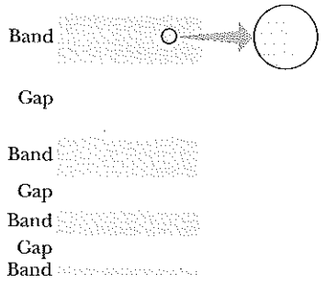


FIGURE 46-3 An idealized representation of the band–gap pattern for the energy levels of a solid. As the magnified view of the upper band suggests, each band consists of a very large number of energy levels that lie very close together. The levels of this idealized plot are, as yet, unoccupied by electrons.

way, electrons in individual atoms cannot possess energies that lie between the discrete allowed levels of the atom.

Note in Fig. 46-3 that the bands that lie lower in energy are narrower than those that lie higher. This is because these low-energy bands correspond to levels in the isolated atom that are occupied by electrons that spend most of their time deep within the electron cloud of the atom. The wave functions of these core electrons thus do not overlap as much as do the wave functions of the outer, or valence, electrons and for that reason the splitting of the levels —although it must occur—is not as great as it is for the levels normally occupied by the outer electrons.

Now that we have established the pattern of levels for a solid, we are ready to consider how these levels are filled with electrons. We shall see how this will lead us in a convincing way to the answers to the question that we raised at the end of Section 46-2.

46-4 INSULATORS

The feature that defines an *insulator* is that, as Fig. 46-4 shows, the highest occupied level coincides with the top of a band. In addition, this band must be separated from the unoccupied band above it by a substantial energy gap E_g. For diamond, $E_g =$ 5.4 eV, a value about 140 times larger than the average thermal energy of a free particle at room temperature.

By definition, an insulator is a solid through which electrons cannot flow as a directed drift current. Let us see why. If you apply an electric field **E** to an insulator, it will exert a force $-e\mathbf{E}$ on each electron. Classically, this force will cause the electron to increase the component of its velocity in the direction $-\mathbf{E}$, which in turn means that its kinetic energy will change. In quantum terms, if the energy of an electron changes, the electron must move to a different energy level within the solid. In an insulator, however, the Pauli principle prevents the electron from doing so because all other levels within the band into which the electron might move are already occupied. These electrons are in total gridlock. It is as if a child tries to climb a ladder on which other children are standing, one every rung; since there are no vacant rungs, no one can move.

There are plenty of vacant levels in the band above the filled band in Fig. 46-4 but, if an electron is to occupy one of these levels, it must somehow jump across the gap that separates the two bands. It cannot pause at a way station within the gap because all energies in this range are strictly forbidden. In diamond, the gap is simply too wide for any detectable number of electrons to make it to the vacant band, either by the action of an external electric field or by thermal agitation.

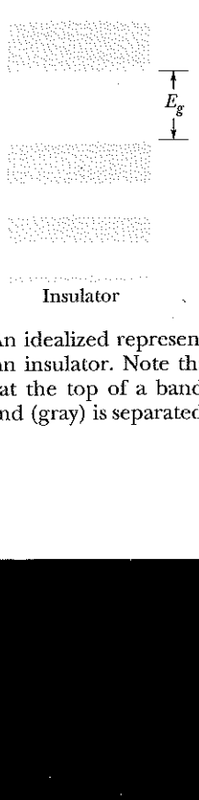


FIGURE 46-4 An idealized representation of the band–gap pattern for an insulator. Note that the highest filled level (gold) lies at the top of a band and that the next highest vacant band (gray) is separated by a relatively large energy gap E_g.

46-5 METALS: QUALITATIVE

The feature that defines a *metal* is that, as Fig. 46-5 shows, the highest occupied level (at the absolute zero of temperature) falls somewhere in the middle of a band. The electrons that occupy this partially filled band are the valence electrons of the atoms,

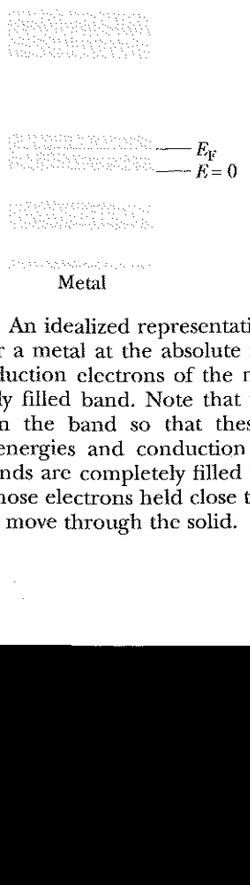


FIGURE 46-5 An idealized representation of the band–gap pattern for a metal at the absolute zero of temperature. The conduction electrons of the metal occupy the highest partially filled band. Note that vacant levels are available within the band so that these electrons can change their energies and conduction can take place. Lower lying bands are completely filled by the core electrons, that is, those electrons held close to the lattice sites and not free to move through the solid.

which, being free to move throughout the solid, become the *conduction electrons* of the solid.

At the absolute zero of temperature, thermal agitation plays no role and all electrons occupy states of the lowest possible energy. We can assume with little error that the potential energy of the conduction electrons remains constant as they move about within the solid. If we set this constant equal to zero —as we are always permitted to do—the total energy E associated with any level is equal to the kinetic energy of the electron that occupies that level.

The level at the bottom of the partially filled band of Fig. 46-5 thus corresponds to $E = 0$. The highest occupied level in this band (at the absolute zero of temperature) is called the **Fermi level** and the energy corresponding to it is called the **Fermi energy** E_F; for copper, $E_F = 7.0$ eV. The electron speed corresponding to the Fermi energy is called the **Fermi speed** v_F; for copper, $v_F = 1.6 \times 10^6$ m/s.

A glance at Fig. 46-5 should be enough to shatter the popular misconception that all motion ceases at the absolute zero of temperature. We see that, entirely because of Pauli's exclusion principle, the electrons are stacked up in the partially filled band of Fig. 46-5 with energies that range from zero up to the Fermi energy. The *average* kinetic energy for the electrons in this band for copper is about 4.2 eV. By comparison, the average translational kinetic energy of a molecule of an ideal gas at room temperature is only 0.025 eV. The conduction electrons in a metal have plenty of energy at absolute zero!

Conditions for $T > 0$

What happens to the electron distribution of Fig. 46-5 as we raise the temperature above absolute zero? The short answer is that not very much happens, although the little that does is very important. It is clear that electrons in bands below the partially filled band of Fig. 46-5 cannot be affected by thermal agitation; they are all gridlocked.

Only electrons close to the Fermi energy find vacant levels above them and it is only these electrons that are free to be boosted to higher levels by thermal agitation. Even at $T = 1000$ K, a temperature at which a metal sample would glow brightly in a dark room, the distribution of electrons among the available levels does not differ very much from the distribution at the absolute zero.

Let us see why. The quantity kT, where k is the Boltzmann constant (8.62×10^{-5} eV/K), is a convenient measure of the energy that may be given to an electron by the random thermal jiggling of the lattice. At $T = 1000$ K, $kT = 0.086$ eV; no electron can hope to have its energy changed by more than a few times this relatively small amount by thermal agitation alone. All the "action" takes place for electrons whose energies are close to the Fermi energy. It has been said, somewhat poetically, that thermal agitation normally causes only ripples on the surface of the Fermi sea; the vast depths of that sea lie undisturbed.

Electrical Conduction in a Metal

If you apply an electric field $\mathbf{E}$ to a metal, the field exerts a force $-e\mathbf{E}$ on each electron. This force, during time Δt, causes every conduction electron in the metal to acquire a velocity increment $\Delta\mathbf{v}$ in the direction of $-\mathbf{E}$. This change in velocity requires that the electrons change their energies, but there are vacant levels available so that these rearrangements can be made. To return to our previous metaphor, there are now vacant rungs on the upper half of the ladder.

The velocities of the individual conduction electrons do not increase without limit, however, because of collisions associated with the thermal vibrations of the lattice. Thus, after a certain time τ, called the *relaxation time*, the drift velocity of the conduction electrons settles down to a constant limiting value, which we associate with the constant current that is set up by the applied electric field. Note that, although *all* the conduction electrons contribute to the current, only electrons close to the Fermi energy are able to make collisions and thus play their role in establishing the limiting value of the drift velocity. It is only these electrons that have ample vacant levels nearby into which they can move after they have experienced a scattering event.

In Section 28-6, we presented the following equation for the resistivity of a metallic conductor,

$$\rho = \frac{m}{ne^2\tau}, \tag{46-1}$$

in which m is the mass of the electron, $-e$ is its charge, and n is the number density of the charge carriers, that is, the number of conduction electrons per unit volume. Although we derived this equation on a classical basis, it holds true when the quantization of the electron energy is taken into account. The quantity τ is the relaxation time to which we referred in the preceding paragraph.

SAMPLE PROBLEM 46-1

How many conduction electrons are there in a copper cube 1.00 cm on edge?

SOLUTION In copper, there is one conduction electron per atom so that N is given by

$$N = na^3, \tag{46-2}$$

in which n is the number of atoms per unit volume and a (= 1.00 cm) is the length of the cube edge. We can find n from

$$n = \frac{N_A d}{A}$$

in which N_A is the Avogadro constant, A is the molar mass of copper, and d is the density of copper. If we substitute values for these quantities, we find

$$n = \frac{(6.02 \times 10^{23}\ \text{atoms/mol})(8900\ \text{kg/m}^3)}{0.06357\ \text{kg/mol}}$$

$$= 8.43 \times 10^{28}\ \text{atoms/m}^3$$

$$= 8.43 \times 10^{28}\ \text{electrons/m}^3.$$

From Eq. 46-2 we then have

$$N = na^3 = (8.43 \times 10^{28}\ \text{electrons/m}^3)(1.00 \times 10^{-2}\ \text{m})^3$$

$$= 8.43 \times 10^{22}\ \text{electrons.} \qquad \text{(Answer)}$$

The Pauli exclusion principle requires that each of these electrons occupy a different quantum state. These states are distributed over an energy interval of only 7.0 eV (the Fermi energy) so that the average spacing between them is very small indeed.

Not all these states have different energies. Consider, for example, an electron moving along the x axis with speed v. It has the same energy as an electron moving with this same speed in any other direction in the solid but, because its motion is different, it is in a different quantum state, described by a different wave function.

SAMPLE PROBLEM 46-2

a. What is the speed of a conduction electron in copper with a kinetic energy equal to the Fermi energy (= 7.0 eV)?

SOLUTION The total energy E of the conduction electrons is all kinetic and we can write, if $E = E_F$,

$$E_F = \tfrac{1}{2}mv_F^2,$$

in which v_F is the Fermi speed. Solving for v_F yields

$$v_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{(2)(7.0\ \text{eV})(1.6 \times 10^{-19}\ \text{J/eV})}{9.11 \times 10^{-31}\ \text{kg}}}$$

$$= 1.6 \times 10^6\ \text{m/s.} \qquad \text{(Answer)}$$

You must not confuse this speed with the *drift speed* of the conduction electrons, which is typically 10^{-5} m/s and is thus smaller by about a factor of 10^{11}. As we explained more fully in Section 28-6, the drift speed is the average speed at which electrons actually drift through a conductor when an electric field is applied; the Fermi speed is the average speed of these electrons between collisions.

b. What is the average time τ between collisions for the conduction electrons in copper? The resistivity of copper at room temperature is $1.7 \times 10^{-8}\ \Omega \cdot \text{m}$.

SOLUTION Solving Eq. 46-1 for τ yields

$$\tau = \frac{m}{ne^2\rho}$$

$$= \frac{9.11 \times 10^{-31}\ \text{kg}}{8.43 \times 10^{28}\ \text{m}^{-3}} \times \frac{1}{(1.6 \times 10^{-19}\ \text{C})^2\,(1.7 \times 10^{-8}\ \Omega \cdot \text{m})}$$

$$= 2.5 \times 10^{-14}\ \text{s.} \qquad \text{(Answer)}$$

c. What mean free path λ may be calculated from the results of (a) and (b) above?

SOLUTION We have

$$\lambda = v_F\tau = (1.6 \times 10^6\ \text{m/s})(2.5 \times 10^{-14}\ \text{s})$$

$$= 4.0 \times 10^{-8}\ \text{m} = 40\ \text{nm.} \qquad \text{(Answer)}$$

In the copper lattice the centers of neighboring atoms are 0.26 nm apart. Thus a typical conduction electron can move a substantial distance, about 150 interatomic distances, through a copper lattice at room temperature before making a collision.

46-6 METALS: QUANTITATIVE

Now let us look at the conduction of electricity in a metal quantitatively, under several headings.

Counting the Quantum States

We start by counting the number of distinct quantum states in the partially filled band of Fig. 46-5. We

cannot possibly deal with this vast number of states one at a time; we must use statistical methods. Instead of asking, "What is the energy of this state?" we must ask, "How many states (per unit volume) have energies that lie in the energy range E to $E + dE$?" This number can be written as $n(E)\,dE$, where $n(E)$ is called the **density of states.**

If we assume that the conduction electrons move in a region of constant potential, $n(E)$ can be shown to be given by

$$n(E) = \frac{8\sqrt{2}\pi m^{3/2}}{h^3} E^{1/2} \quad \text{(density of states)}. \quad (46\text{-}3)$$

(See Eisberg and Resnick, *Quantum Physics,* second edition, 1985, Wiley, Section 13-5.) Figure 46-6*a* is a plot of Eq. 46-3. Note that there is nothing in this equation that depends on the shape or size of the sample or on the material of which it is made.

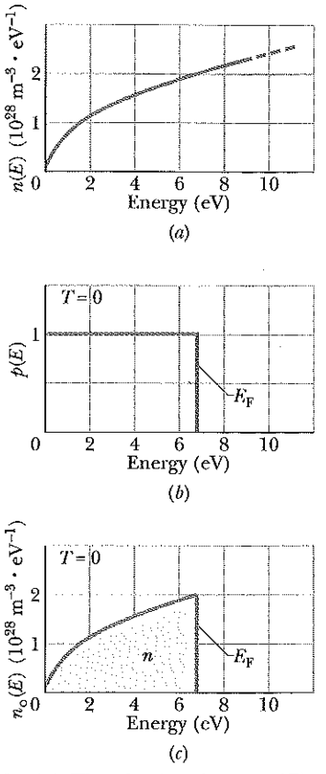


FIGURE 46-6 (*a*) The *density of states* $n(E)$ plotted as a function of energy. (*b*) The *probability function* $p(E)$ plotted as a function of energy at $T = 0$. (*c*) The *density of occupied states* $n_o(E)$, formed by multiplying the curves in (*a*) and (*b*), plotted as a function of energy. Note that all states whose energies lie below the Fermi energy are occupied but all states above that energy are vacant.

Filling the States at $T = 0$

Equation 46-3 tells us how the *unoccupied* states are distributed in energy. We must now weight each energy with a factor $p(E)$, called the **probability function,** that gives us the probability that a state with that energy will actually be occupied. At the absolute zero of temperature, all states with energies less than the Fermi energy are filled and all states with energies greater than that energy are vacant. The probability function in this case must be the simple rectangle shown in Fig. 46-6*b*, in which unity corresponds to a certainty that the state will be occupied and zero corresponds to a certainty that the state will *not* be occupied.

The product of these two factors gives us $n_o(E)$, the density of *occupied* states. Thus

$$n_o(E) = n(E)\,p(E). \quad (46\text{-}4)$$

Figure 46-6*c* is a plot of this product.

We can find the Fermi energy for a metal by adding up (integrating) the number of occupied states in Fig. 46-6*c* between $E = 0$ and $E = E_F$. The result must equal n, the number of conduction electrons per unit volume for the metal. In equation form we have

$$n = \int_0^{E_F} n(E)\,dE. \quad (46\text{-}5)$$

Note that n is represented by the beige area shown in Fig. 46-6*c*.

Calculating the Fermi Energy

If we substitute Eq. 46-3 into Eq. 46-5, we find

$$n = \frac{8\sqrt{2}\pi m^{3/2}}{h^3} \int_0^{E_F} E^{1/2}\,dE = \left(\frac{8\sqrt{2}\pi m^{3/2}}{h^3}\right)\left(\frac{2E_F^{3/2}}{3}\right).$$

Solving for E_F leads to

$$E_F = \left(\frac{3}{16\sqrt{2}\pi}\right)^{2/3} \frac{h^2}{m} n^{2/3} = \frac{0.121h^2}{m} n^{2/3}. \quad (46\text{-}6)$$

Thus the Fermi energy can be calculated once n, the number of conduction electrons per unit volume, is known.

Filling the States for $T > 0$

It can be shown that the probability function for $T > 0$ is given by

$$p(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \quad \text{(probability function)}, \tag{46-7}$$

in which E_F is the Fermi energy and k is the Boltzmann constant. (See Eisberg and Resnick, *Quantum Physics*, second edition, 1985, Wiley, Section 11-4.)

Note that as $T \to 0$, the exponent $(E - E_F)/kT$ in Eq. 46-7 approaches $-\infty$ if $E < E_F$ and $+\infty$ if $E > E_F$. In the first case we have $p(E) = 1$ and in the second $p(E) = 0$. Thus, at $T = 0$, Eq. 46-7 correctly yields the rectangular form shown in Fig. 46-6*b*. Equation 46-7 also shows us that the important quantity is not the energy E but rather $E - E_F$, the energy interval between E and the Fermi energy.

Figure 46-7*b* shows the probability function for $T = 1000$ K, calculated from Eq. 46-7. Note how little it differs from the rectangular form of Fig. 46-6*b*. Figure 46-7*c*, found by multiplying Figs. 46-7*a* and 46-7*b*, shows the density of occupied states for $T = 1000$ K. Note how little that differs from Fig. 46-6*c*, the distribution at $T = 0$.

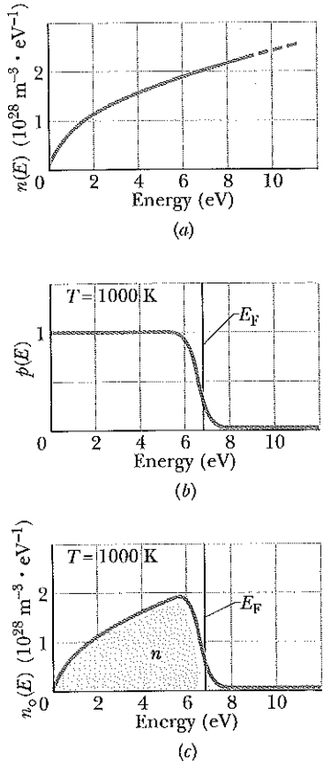


FIGURE 46-7 (*a*) The *density of states* $n(E)$ plotted as a function of energy; this plot is the same as that of Fig. 46-6*a*. (*b*) The *probability function* $p(E)$ plotted as a function of energy at $T = 1000$ K; note how little this plot differs from that of Fig. 46-6*b*. (*c*) The *density of occupied states* $n_o(E)$, formed by multiplying the curves in (*a*) and (*b*), plotted as a function of energy; note how little this plot differs from that of Fig. 46-6*c*. (This plot is idealized in that it assumes that the conduction electrons move in a region of constant potential. Measured density of states plots do not have this simple shape.)

SAMPLE PROBLEM 46-3

A cube of copper is 1.00 cm on an edge. How many quantum states lie in the energy interval between $E = 5.00$ eV and $E = 5.01$ eV?

SOLUTION These energy limits are so close together that we can say that the answer is

$$N = n(E)\,\Delta E\,V, \tag{46-8}$$

where $E = 5.00$ eV, $\Delta E = 0.01$ eV, and V is the volume of the cube. From Eq. 46-3 we have

$$n(E) = \frac{8\sqrt{2}\pi m^{3/2}}{h^3} E^{1/2}$$

$$= (8\sqrt{2}\pi)(9.11 \times 10^{-31}\ \text{kg})^{3/2} \times \frac{(5.00\ \text{eV})^{1/2}(1.60 \times 10^{-19}\ \text{J/eV})^{1/2}}{(6.63 \times 10^{-34}\ \text{J·s})^3}$$

$$= 9.48 \times 10^{46}\ \text{m}^{-3}\,\text{J}^{-1} = 1.52 \times 10^{28}\ \text{m}^{-3}\cdot\text{eV}^{-1}.$$

From Eq. 46-8 we have, putting $V = a^3$, where a is the cube edge,

$$N = n(E)\,\Delta E\,a^3$$

$$= (1.52 \times 10^{28}\ \text{m}^{-3}\ \text{eV}^{-1})(0.01\ \text{eV})(1 \times 10^{-2}\ \text{m})^3$$

$$= 1.52 \times 10^{20}. \quad \text{(Answer)}$$

SAMPLE PROBLEM 46-4

a. What is the probability that a state whose energy is 0.10 eV above the Fermi energy will be occupied? Assume a temperature of 800 K.

SOLUTION We can find $p(E)$ from Eq. 46-7. Let us first calculate the (dimensionless) exponent in that equation:

$$\frac{E - E_F}{kT} = \frac{0.10 \text{ eV}}{(8.62 \times 10^{-5} \text{ eV/K})(800 \text{ K})} = 1.45.$$

Inserting this exponent into Eq. 46-7 yields

$$p = \frac{1}{e^{1.45} + 1} = 0.19 \text{ or } 19\%. \quad \text{(Answer)}$$

b. What is the probability of occupancy for a state that is 0.10 eV *below* the Fermi energy?

SOLUTION The exponent in Eq. 46-7 has the same numerical value as above but is now negative. Thus from this equation

$$p = \frac{1}{e^{-1.45} + 1} = 0.81 \text{ or } 81\%. \quad \text{(Answer)}$$

For states whose energies lie below the Fermi energy we are often more interested in the probability that the state is *not* occupied. This is, of course, just $1 - p$, or 19% in the present case. An unfilled state in an energy range in which most of the states are filled is called a *hole*. We shall see later that this is a very useful concept.

c. What is the probability of occupancy for a state whose energy is equal to the Fermi energy?

SOLUTION For $E = E_F$ the exponent in Eq. 46-7 is zero and that equation becomes

$$p = \frac{1}{e^0 + 1} = \frac{1}{1 + 1} = 0.50 \text{ or } 50\%. \quad \text{(Answer)}$$

This result does not depend on the temperature. We can, in fact, define the Fermi energy for a metal to be that energy for which the probability of occupancy *at any temperature* is 50%.

46-7 SEMICONDUCTORS

As a comparison of Fig. 46-8 with Fig. 46-4 shows, a semiconductor is like an insulator in that its uppermost filled level (at the absolute zero of temperature) lies at the top of a band. A semiconductor differs from an insulator, however, in that the gap between this filled band and the next vacant band above it is much smaller than for an insulator, so that there is a real possibility for electrons to "jump the gap" into this empty band by thermal agitation. For semiconducting materials, the highest filled band is called the **valence band** because the electrons that occupy it are the valence electrons of the isolated atom. The band above the valence band, which is vacant at $T = 0$, is called the **conduction band.**

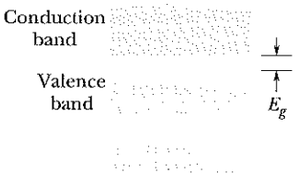


FIGURE 46-8 An idealized representation of the band–gap pattern for a semiconductor such as silicon. It resembles the pattern for an insulator (see Fig. 46-4) except that here the gap between the valence band and the conduction band is much smaller.

The distinction between an insulator and a semiconductor is qualitative, depending as it does on the width of the energy gap. However, there is no doubt that diamond ($E_g = 5.4$ eV) is an insulator and silicon ($E_g = 1.1$ eV) is a semiconductor. As it happens (see Fig. 46-1) these two substances have the same crystal structure.

The revolution in microelectronics that has so influenced our lives is based on semiconductors (see Fig. 46-9), so we would do well to learn more about them. Table 46-1 compares some electrical properties of silicon, our prototype semiconductor, and copper, our prototype conductor. Let us look carefully at this table, one row at a time.

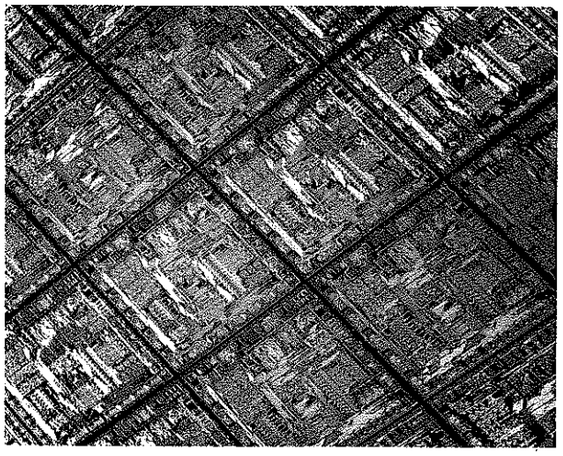


FIGURE 46-9 A photograph of an enlarged section of an integrated circuit.

The Density of Charge Carriers, n

Copper has many more charge carriers than does silicon, by a factor of about 10^{13}. For copper the carriers are the conduction electrons, present in the number of one per atom. At room temperature, to which Table 46-1 refers, charge carriers in silicon arise only because, at thermal equilibrium, thermal agitation has caused a certain (very small) number of electrons to be raised to the conduction band, leaving an equal number of vacant states (holes) in the valence band.

The holes in the valence band of a semiconductor also serve effectively as charge carriers because they permit a certain freedom of movement to the electrons in that band. If an electric field is set up in a semiconductor, the electrons in the valence band, being negatively charged, will drift opposite the direction of **E**. This causes the holes to drift in the direction of **E**. That is, the holes behave like particles carrying a charge $+e$ and, in all that follows, that is exactly how we shall regard them. Conduction by holes is an important fact of life for semiconductors.

If the concept of a migrating hole seems confusing to you, think of a vacant slot in a parking lot that is otherwise filled with cars. If one of these cars moves into the slot, it fills the slot but creates a new vacant slot in the place it just left. This vacancy, in turn, can be filled by another car, and so on. As the cars move around in this way, we can focus attention on the single migrating vacant slot as it wanders over the lot.

The Resistivity, ρ

At room temperature the resistivity of silicon is considerably higher than that of copper, by a factor of about 10^{11}. For both elements, the resistivity is determined by Eq. 46-1. The vast difference in resistivity between copper and silicon can be accounted for by the vast difference in n, the density of charge carriers. (The mean collision time τ will also be different for copper and for silicon but the effect of this on the resistivity is swamped by the enormous difference in n.)

The Temperature Coefficient of Resistivity, α

This quantity (see Eq. 28-16) is the fractional change in resistivity per unit change in temperature, or

$$\alpha = \frac{1}{\rho}\frac{d\rho}{dT}.$$

The resistivity of copper and other metals *increases* with temperature ($d\rho/dT > 0$). This happens because collisions occur more frequently the higher the temperature, thus reducing τ in Eq. 46-1. For metals, the density of charge carriers n in that equation is independent of temperature.

On the other hand, the resistivity of silicon (and other semiconductors) *decreases* with temperature ($d\rho/dT < 0$). This happens because the density of charge carriers n in Eq. 46-1 increases rapidly with temperature. The decrease in τ mentioned before for metals also occurs for semiconductors but its effect on the resistivity is swamped by the very rapid increase in the density of charge carriers.

46-8 DOPING

The versatility of semiconductors can be marvelously improved by introducing a small number of suitable

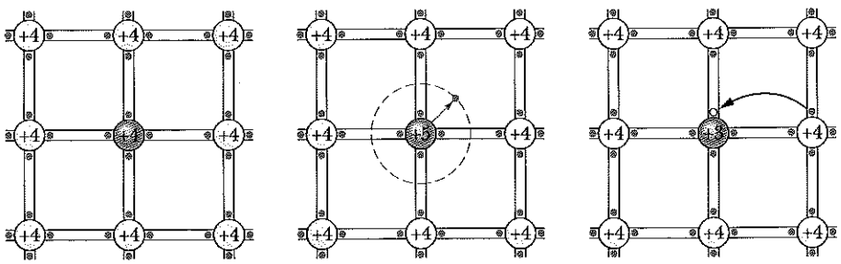


FIGURE 46-10 (*a*) A two-dimensional representation of a silicon lattice. Each silicon ion (core charge $= +4e$) is bonded to each of its four nearest neighbors by a shared two-electron bond. The red dots show these valence electrons. (*b*) A phosphorus atom (valence = 5) is substituted for the central silicon atom, creating a donor site. (*c*) An aluminum atom (valence = 3) is substituted for the central silicon atom, creating an acceptor site.

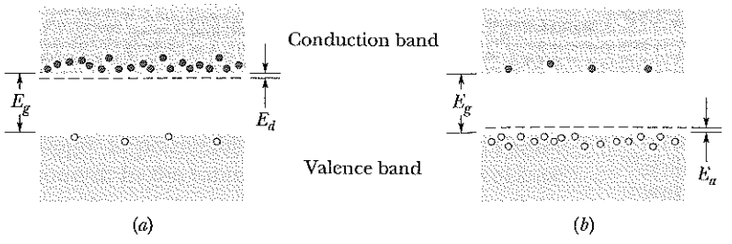


FIGURE 46-11 (*a*) An *n*-type semiconductor, showing the donor level that has contributed most of the electrons (majority carriers) in the conduction band. The small number of holes (minority carriers) in the valence band is also shown. (*b*) A *p*-type semiconductor, showing the acceptor level that has contributed most of the holes (majority carriers) in the valence band. The small number of electrons (minority carriers) in the conduction band is also shown.

replacement atoms (it seems pejorative to call them impurities) into the semiconductor lattice, a process called **doping.** Essentially all practical semiconducting devices today are based on doped material. They are of two varieties, called *n-type* and *p-type;* we discuss each in turn.

n-Type Semiconductors

Figure 46-10*a* is a "flattened out" representation of a lattice of pure silicon; compare Fig. 46-1*a*. Each silicon atom forms a two-electron covalent bond with each of its four nearest neighbors, the electrons involved in the bonding making up the valence band of the sample. In Fig. 46-10*b* one of the silicon atoms (valence = 4) has been replaced by an atom of phosphorus (valence = 5). As Fig. 46-10*b* suggests, the "extra" electron is loosely bound to the phosphorus ion core because it is not involved in covalent bonds to neighboring ions. It is far easier for *this* electron to be thermally excited into the conduction band than it is for one of the silicon valence electrons to be so excited.

The phosphorus atom is called a **donor** atom because it so readily *donates* an electron to the conduction band. The "extra" electron in Fig. 46-10*b* can be said to lie in a localized *donor level,* as Fig. 46-11*a* shows. This level is separated from the bottom of the conduction band by an energy gap E_d, where $E_d \ll E_g$. By controlling the concentration of donor atoms it is possible to increase greatly the density of electrons in the conduction band.

Semiconductors doped with donor atoms are called *n-type* semiconductors, the *"n"* standing for "negative" because the negative charge carriers greatly outnumber the positive charge carriers. The former, called the *majority carriers,* are the electrons in the conduction band. The latter, called the *minority carriers,* are the holes in the valence band.

TABLE 46-2
PROPERTIES OF TWO DOPED SEMICONDUCTORS

Matrix material	Silicon	Silicon
Dopant	Phosphorus	Aluminum
Type of dopant	Donor	Acceptor
Type of semiconductor	*n*-Type	*p*-Type
Dopant valence	5 (=4 + 1)	3 (=4 − 1)
Dopant energy gap	45 meV	57 meV
Majority carriers	Electrons	Holes
Minority carriers	Holes	Electrons
Dopant ion core charge	$+e$	$-e$

p-Type Semiconductors

Figure 46-10*c* shows a silicon lattice in which a silicon atom (valence = 4) has been replaced by an aluminum atom (valence = 3). Now there is a "missing" electron and it is easy for the aluminum ion to "steal" a valence electron from a nearby silicon atom, thus creating a hole in the valence band.

The aluminum atom is called an **acceptor** atom because it so readily *accepts* an electron from the valence band. The electron so accepted moves into a localized *acceptor level,* as Fig. 46-11*b* shows. This level is separated from the top of the valence band by an energy gap $E_a \ll E_g$. By controlling the concentration of acceptor atoms it is possible to greatly increase the number of holes in the valence band.

Semiconductors doped with acceptor atoms are called *p-type* semiconductors, the *"p"* standing for "positive" because the positive charge carriers in this case greatly outnumber the negative carriers. In *p*-type semiconductors the majority carriers are the holes in the valence band and the minority carriers are the electrons in the conduction band.

Table 46-2 summarizes the properties of a typical *n*-type and a typical *p*-type semiconductor. Note particularly that the donor and acceptor ion cores, although they are charged, are not charge *carriers* because, at normal temperatures, they remain fixed in their lattice sites.

SAMPLE PROBLEM 46-5

The number density of conduction electrons in pure silicon at room temperature is about $10^{16}\ m^{-3}$. Assume that, by doping the lattice with phosphorus, you want to increase this number by a factor of a million (10^6). What fraction of the silicon atoms must you replace with phosphorus atoms? (Assume that, at room temperature, the thermal agitation is effective enough so that essentially every phosphorus atom donates its "extra" electron to the conduction band.)

SOLUTION The density of the phosphorus atoms must be about $(10^{16}\ m^{-3})(10^6)$, or about $10^{22}\ m^{-3}$. The density of silicon atoms in a pure silicon lattice may be found from

$$n_{Si} = \frac{N_A d}{A},$$

in which N_A is the Avogadro constant, d is the density of silicon, and A is the molar mass of silicon; from Appendix D we find that $d = 2330\ kg/m^3$ and $A = 28.1\ g/mol$. Substituting yields

$$n_{Si} = \frac{(6.02 \times 10^{23}\ mol^{-1})(2330\ kg/m^3)}{0.0281\ kg/mol}$$

$$= 5 \times 10^{28}\ m^{-3}.$$

The ratio of these two number densities is the quantity we are looking for. Thus

$$\frac{n_{Si}}{n_P} = \frac{5 \times 10^{28}\ m^{-3}}{10^{22}\ m^{-3}} = 5 \times 10^6. \quad \text{(Answer)}$$

We see that if only *one silicon atom in five million* is replaced by a phosphorus atom, the number of electrons in the conduction band will be increased by a factor of 10^6.

How can such a tiny admixture of phosphorus atoms have such a big effect? The answer is that, for pure silicon at room temperature, there were not many conduction electrons to start with. The density of conduction electrons was $10^{16}\ m^{-3}$ before doping and $10^{22}\ m^{-3}$ after doping. For copper, however, the conduction electron density (see Table 46-1) is about $10^{29}\ m^{-3}$. Thus, even *after* doping, the conduction electron density of silicon remains much less than that of a typical metal such as copper.

46-9 THE *p-n* JUNCTION

Pass a hypothetical plane across a rod of pure silicon. Dope the rod on one side of the plane with donor atoms (thus creating *n*-type material) and on the other side with acceptor atoms (thus creating *p*-type material). You have just made a ***p-n* junction;** it is at the heart of essentially all semiconducting devices.* Figure 46-12*a* represents a *p-n* junction at the imagined moment of its creation. Let us first discuss the motions of the majority carriers, which are electrons in the *n*-type material and holes in the *p*-type material.

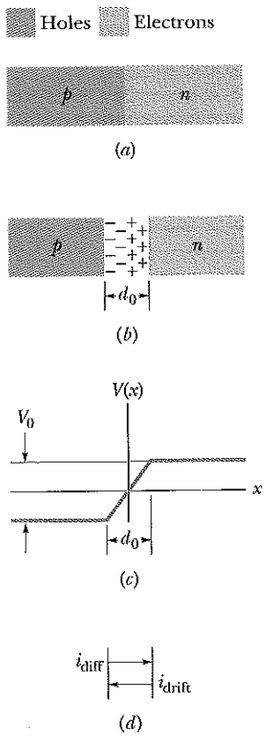


FIGURE 46-12 (*a*) A *p-n* junction at the imagined moment of its creation. Only the majority carriers are shown. (*b*) Diffusion of majority carriers across the junction plane causes a space charge of fixed donor and acceptor ions to appear. (*c*) The space charge establishes a contact potential difference V_0 across the junction plane. (*d*) In equilibrium, the diffusion of majority carriers across the junction plane is just balanced by the drift of minority carriers in the opposite direction.

*In practice, to make a *p-n* junction one usually starts with, say, *n*-type material and then diffuses acceptor atoms into the solid sample at high temperature, overcompensating the donor atoms to a certain (controllable) depth below the surface.

Motions of the Majority Carriers

Electrons close to the junction plane will tend to diffuse across it (from right to left in Fig. 46-12), for much the same reason that gas molecules will diffuse through a permeable membrane into a vacuum beyond it. In the same way, holes will tend to diffuse across the junction plane from left to right. Both motions contribute to a *diffusion current* i_{diff}, directed from left to right as in Fig. 46-12*d*.

Recall that *n*-type material is studded throughout with donor ions, fixed firmly in their lattice sites. Normally, the positive charges of these ions are compensated electrically by the majority carriers, which are electrons. When an electron diffuses from the *n*-type material and through the junction plane, however, it "uncovers" one of these donor ions, thus introducing a fixed positive charge in the *n*-type material. When this diffusing electron arrives on the other side of the barrier, it quickly finds a hole and combines with it,* thus neutralizing one of the positively charged acceptor ions that are sprinkled throughout the *p*-type material, resulting in a fixed negative charge in the *p*-type material.

Convince yourself that a hole diffusing through the barrier from left to right has exactly the same end result. Thus a region of fixed positive charge builds up on one side of the barrier and a region of fixed negative charge builds up on the other (Fig. 46-12*b*). These two regions form the so-called **depletion zone;** the fixed charges are said to be positive and negative **space charge.**

The space charge causes a *contact potential difference* to build up across the junction, as Fig. 46-12*c* shows. This potential difference is such that it serves as a barrier to limit further diffusion of both electrons and holes across the junction plane. An electron at the junction plane, for example, would be repelled back to its *n*-type home by the negative space charge in the *p*-type material that faces it across the plane. To complete the picture, let us turn our attention to the minority carriers.

Motions of the Minority Carriers

As Fig. 46-11*a* and Table 46-2 show, although the majority carriers in *n*-type material are electrons, there are nevertheless also a few holes, the minority carriers. Likewise in *p*-type material, although the majority carriers are holes, there are also a few conduction electrons.

Although the potential difference in Fig. 46-12*c* acts to retard the motions of the majority carriers—being a barrier for them—it is a downhill trip for the minority carriers, be they electrons or holes. When, by thermal agitation, an electron close to the junction plane is raised from the valence band to the conduction band of the *p*-type material in Fig. 46-12*a*, the contact potential difference causes it to drift steadily from left to right across the junction plane. Similarly, if a hole is created in the *n*-type material, it too drifts across to the other side. The space-charge region shown in Fig. 46-12*b* is effectively swept free of charge carriers by this process and, for that reason, we call it the *depletion* zone. The current represented by the motions of the minority carriers, called the *drift current* i_{drift}, is in the opposite direction to the diffusion current and just compensates it at equilibrium, as Fig. 46-12*d* shows.

Thus, at equilibrium, a *p-n* junction resting on a shelf develops a contact potential difference V_0 between its ends. The diffusion current i_{diff} that moves through the junction plane from the direction *p* to *n* is just balanced by a drift current i_{drift} that moves in the opposite direction.

46-10 THE DIODE RECTIFIER

A *p-n* junction is basically a two-terminal rectifier. If you connect this *diode rectifier* across the terminals of a battery, the current in the circuit will be very much smaller for one polarity of the battery connection than for the other, as Fig. 46-13 shows.

Figure 46-14 shows one of many applications of a diode rectifier. A sine wave input potential generates a half-wave output potential, the diode rectifier acting as essentially a short circuit (a closed switch) for one polarity of the input potential and as essentially an open circuit (an open switch) for the other. An ideal diode rectifier, in fact, has only these two modes of operation. It is either ON (zero resistance) or OFF (infinite resistance).

Figure 46-14 displays the conventional symbol for a diode rectifier. The arrowhead corresponds to the *p*-type terminal of the device and points in the direction of "easy" conventional current flow. That is, the diode is ON when the terminal with the ar-

*An "electron combines with a hole" when the electron drops from the conduction band to the valence band, filling a vacancy in that band.

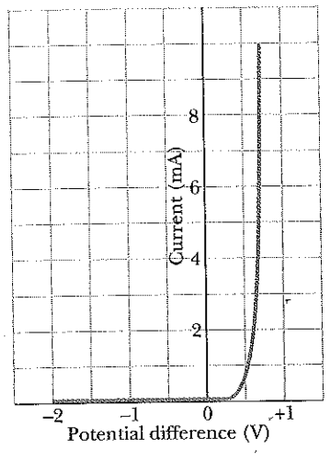


FIGURE 46-13 A current–voltage plot for a junction diode, showing that it is highly conducting in the forward direction and essentially nonconducting in the reverse direction.

rowhead is (sufficiently) positive with respect to the other terminal.

Figure 46-15 shows details of the two connections. In Fig. 46-15*a*—the back-bias arrangement—the battery emf simply *adds* to the contact potential difference, thus increasing the height of the barrier that the majority carriers must surmount. Fewer of them can do so and, as a result, the diffusion current decreases markedly.

The drift current, however, senses no barrier and thus is independent of the magnitude or direction of the external potential. The nice current balance that existed at zero bias (see Fig. 46-12*d*) is thus upset and, as shown in Fig. 46-15*a*, a very small net back-current i_B appears in the circuit.

Another effect of back-bias is to widen the depletion zone, as a comparison of Figs. 46-12*b* and 46-15*a* shows. Because the depletion zone contains

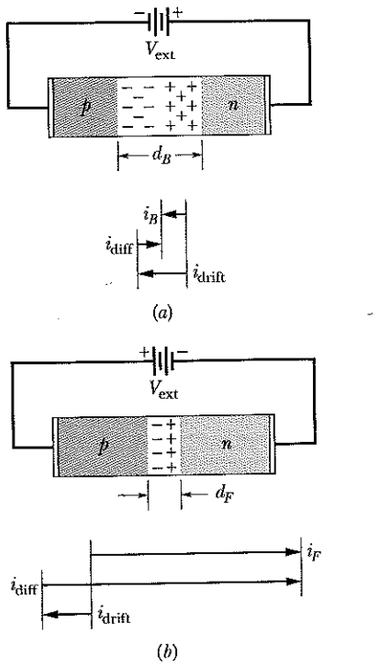


FIGURE 46-15 (*a*) The back-bias connection of a *p*-*n* junction, showing the wide depletion zone and the corresponding small back current. (*b*) The forward-bias connection, showing the narrowing of the depletion zone and the large forward current.

very few charge carriers, it is a region of high resistivity. Thus its substantially increased width means a substantially increased resistance, consistent with the small value of the back-bias current.

Figure 46-15*b* shows the forward-bias connection, the positive terminal of the battery being connected to the *p*-type end of the *p*-*n* junction. Here the applied emf *subtracts* from the contact potential difference, the diffusion current *rises* substantially,

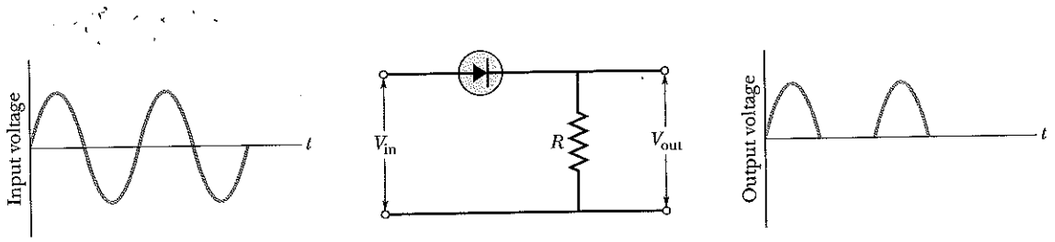


FIGURE 46-14 A *p*-*n* junction diode is connected as a rectifier. The action of the circuit is to pass the positive half of the input wave form but to suppress the negative half. The average potential of the input wave form is zero; that for the output wave form is positive.

and a relatively *large* net forward current i_F results. The depletion zone *narrows*, its low resistance being consistent with the large current i_F.

46-11 THE LIGHT-EMITTING DIODE (LED)

We are all familiar with the brightly colored numbers that flash at us from cash registers and gasoline pumps. In nearly all cases this light is emitted from an assembly of *p-n* junctions operating as **light-emitting diodes** (LEDs).

Figure 46-16*a* shows the familiar seven-segment display from which the numbers are formed. Figure 46-16*b* shows that each element of this display is the end of a flat plastic lens, at the other end of which is a small LED, possibly about 1 mm² in area. Figure 46-16*c* shows a typical circuit, in which the LED is forward-biased.

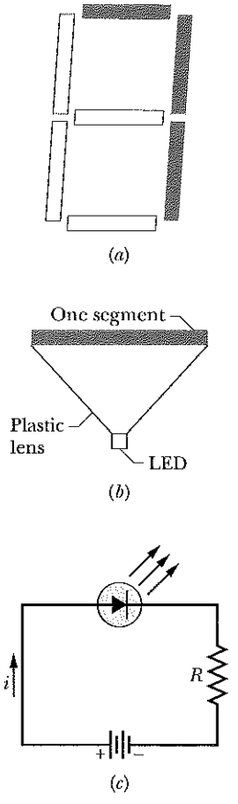


FIGURE 46-16 (*a*) The familiar seven-segment number display, activated to show the number "7." (*b*) One segment of such a display. (*c*) An LED connected to a source of emf.

How can a *p-n* junction emit light? When an electron at the bottom of the conduction band of a semiconductor falls into a hole at the top of the valence band, an energy E_g is released, where E_g is the gap width. What happens to this energy? There are at least two possibilities. It might be transformed into thermal energy of the vibrating lattice and, with high probability, that is exactly what happens in a silicon-based semiconductor.

In some semiconducting materials, however, conditions are such that the emitted energy can also appear as electromagnetic radiation, the wavelength being given by

$$\lambda = \frac{c}{f} = \frac{c}{E_g/h} = \frac{hc}{E_g}. \qquad (46\text{-}9)$$

Commercial LEDs designed for the visible region are usually based on a semiconducting material that is a suitably chosen gallium–arsenic–phosphorus compound. By adjusting the ratio of phosphorus to arsenic, the gap width—and thus the wavelength of the emitted light—can be tailored to suit the need.

A question arises: If light is emitted when an electron falls from the conduction band to the valence band, will not light of that same wavelength be absorbed when an electron moves in the other direction, that is, from the valence band to the conduction band? It will indeed. To avoid having all the emitted photons absorbed, it is necessary to have a great surplus of both electrons and holes present in the material, in much greater numbers than would be generated by thermal agitation in the intrinsic semiconducting material.* These are precisely the conditions that result when majority carriers—be they electrons or holes—are injected across the central plane of a *p-n* junction by the action of an external potential difference. That is why a simple intrinsic semiconductor will not serve as an LED. You need a *p-n* junction! To provide lots of majority carriers (and thus lots of photons), the junction should be heavily doped and strongly forward-biased.

LEDs operating in the infrared are much used in optical communication systems based on optical fibers. The infrared region is chosen because the ab-

*If the surplus of electrons and holes is great enough, there may be a population inversion so that conditions for laser action are set up.

sorption per unit length of such fibers has a well-defined minimum at two different wavelengths.

In another application of the LED, the ends of a suitable *p-n* junction crystal are polished so that a slice of the crystal across the junction plane serves as a laser. Such a device is called a *laser diode;* Fig. 46-17 suggests its tiny scale.

SAMPLE PROBLEM 46-6

An LED is constructed from a *p-n* junction based on a certain Ga–As–P semiconducting material, whose energy gap is 1.9 eV. What is the wavelength of its emitted light?

SOLUTION From Eq. 46-9 we have

$$\lambda = \frac{hc}{E_g} = \frac{(6.63 \times 10^{-34}\ \text{J}\cdot\text{s})(3.00 \times 10^{8}\ \text{m/s})}{(1.9\ \text{eV})(1.60 \times 10^{-19}\ \text{J/eV})}$$

$$= 6.5 \times 10^{-7}\ \text{m} = 650\ \text{nm}. \qquad \text{(Answer)}$$

Light of this wavelength is red.

46-12 THE TRANSISTOR (OPTIONAL)

The devices we have discussed so far have been ·iodes, that is, two-terminal devices. Here we intro-·e a three-terminal device, a **transistor.** As Fig. ·3 suggests, the function of a transistor is to con-current flowing through the device from ter-t· ·· (the **drain**) to terminal S (the **source**) by a·d ·e potential of terminal G (the **gate**). th·s ·· ·y applications, particularly those involv-net·he ·s, we only need to be able to turn the ·be pleti·cen- 46-15 ·exter- intrin- D: You ·rity car- ·n should ed. ·uch used on optical ·ause the ab-

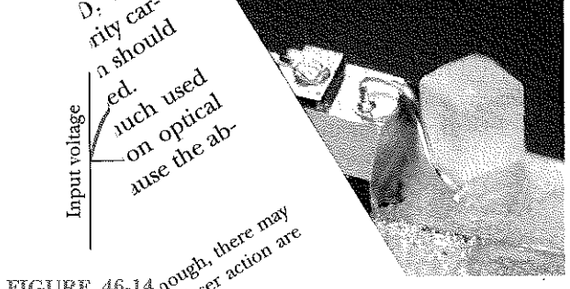


FIGURE 46-14 ·ough, there may rectifier. The acor laser action are half of the input at the AT&T grain of salt.

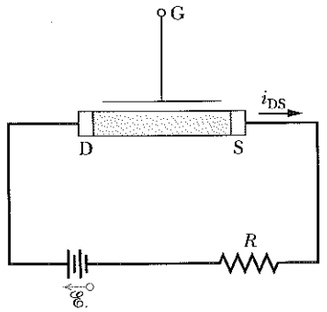


FIGURE 46-18 A representation of a transistor, showing the current i_{DS} moving through the device from the drain D to the source S. The magnitude of the current is controlled by the potential applied to G, the gate terminal.

drain–source current ON (gate open) or OFF (gate closed). One of these conditions corresponds to a "0" and the other to a "1" in the binary arithmetic on which the computer logic is based. We wish further that the gate terminal draw essentially no current from the circuit to which it is attached, thus interfering with its operation as little as possible. In more formal language, we say that we wish the transistor to have a high input impedance.

The central question proves to be: "How can we control the current in a conductor without making direct electric contact with it?" The perhaps surprising answer is that, by using the variable gate potential that is at our disposal, we can change the effective cross-sectional area of the conductor, going even so far as to reduce it to zero.

Figure 46-15 gives a clue. There we see that, by changing the bias potential of the *p-n* junction, we can control the width of the depletion zone. Simply by changing a potential we can effectively transform a conductor (*n*-type or *p*-type material) into a nonconductor (the depletion zone material). We can use the same trick in the transistor, but with a different geometrical arrangement.

Of the several types of transistor that are in common use, we choose to describe the MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor). Figure 46-19 shows its essential features.

A lightly doped *p*-type substrate has imbedded in it two "islands" of heavily doped *n*-type material, forming the drain D and the source S. These terminals are connected by a thin channel of *n*-type material, called the *n*-channel. An insulating layer of silicon dioxide (hence *O*xide in the acronym) is

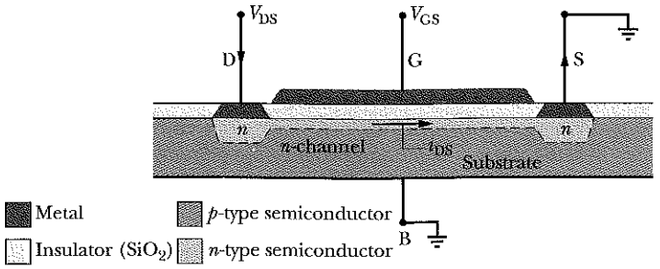


FIGURE 46-19 The construction of a MOSFET. The source S and base B are grounded, and a potential V_{DS} is applied to the drain terminal D. The magnitude of the current is controlled by the gate potential V_{GS}.

deposited on the substrate and penetrated by two metallic contacts (hence *M*etal) at D and S, so that electrical contact can be made with the drain and the source. A thin metallic layer—the gate G—is deposited opposite the *n*-channel. Note that the gate makes no ohmic contact with the transistor proper, being separated from it by the insulating oxide layer. Thus a MOSFET has the desired high input impedance, perhaps as high as $10^{15}\ \Omega$.

Consider first the situation with the source and the substrate grounded, the gate "floating" (that is, not connected electrically to a source of emf), and a positive potential V_{DS} applied to the drain. A drain–source current i_{DS} will be set up, as shown.

The potential difference across the boundary between the *n*-channel and the *p*-type substrate will vary from zero at the source end of the channel to V_{DS} at the drain end. The polarity is such (compare Fig. 46-15*a*) that the *p*-*n* junction that exists at this boundary is back-biased for essentially its full length. A depletion zone will exist at this boundary, increasing in thickness from the source end of the channel to the drain end. For these conditions, the *n*-channel will not have the same cross-sectional area along its length, being invaded by the depletion zone to a greater and greater extent as one proceeds along the channel from the source toward the drain.

The thickness of the depletion zone along its length can be influenced by the potential that we choose to apply to the gate. If we make the gate negative with respect to the source, electrons will be repelled from the *n*-channel into the substrate, thus widening the depletion zone, constricting the channel and decreasing the drain–source current. Alternatively, a positive gate potential will attract electrons into the *n*-channel, narrow the depletion zone, widen the conducting channel, and increase the drain–source current. In this way a small change in the gate potential can generate a substantial change in the drain–source current, much as a valve controls the flow of water through a pipe.

Figure 46-20 shows a MOSFET (note the descriptive symbol) connected into a circuit as an amplifier. The input signal is applied to the gate and the output appears as a varying potential difference across a load resistor.

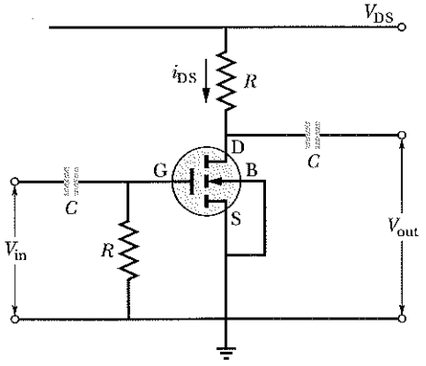


FIGURE 46-20 A MOSFET connected as an amplifier. A time-varying input signal V_{in} generates an amplified output signal V_{out}. The letters G, D, B, and S have the same meanings as in Figs. 46-18 and 46-19.

REVIEW & SUMMARY

Insulators, Metals, and Semiconductors

Quantum mechanics explains why some solids are electrical conductors, others are semiconductors, and why still others are insulators. When atoms are close to each other in a crystal lattice, their atomic energy levels become **bands** of allowed electron energies. In **insulators** the high-

est occupied level coincides with the top of a band; electrons are not able to accept additional kinetic energy from an applied field and so cannot conduct electricity (see Fig. 46-4). In **metals** the highest occupied level (the **Fermi energy**) falls somewhere in the middle of a band (Fig. 46-5 and Sample Problems 46-1 and 46-2). The highest occupied level in a **semiconductor** coincides, at $T = 0$, with the top of a band (the **valence band**) but the energy gap between it and the next highest band (the **conduction band**) is small enough so that charge-carrying electrons can "jump" into the conduction band because of thermal agitation (Fig. 46-8). The higher resistivity and the decrease of resistivity with increasing temperature are both easily explained by this model.

Density of States

Assuming uniform potential, the density of energy states available to electrons in the partially filled band of Fig. 46-5 is

$$n(E) = \frac{8\sqrt{2}\pi m^{3/2}}{h^3} E^{1/2} \qquad \text{(density of states).} \qquad (46\text{-}3)$$

The Fermi Energy

At $T = 0$, electrons fill all the states up to the Fermi energy, with states above the Fermi energy being vacant. The Fermi energy corresponding to Eq. 46-3 is

$$E_F = \frac{0.121h^2}{m} n^{2/3} \qquad \text{(Fermi energy),} \qquad (46\text{-}6)$$

in which n is the number of conduction electrons per unit volume; see Fig. 46-6.

The Probability Function

At temperatures above absolute zero the distribution of occupied states is found by multiplying the density of states by the **probability function**

$$p(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \qquad \text{(probability function)} \qquad (46\text{-}7)$$

as illustrated in Fig. 46-7.

Doping: Donors and Acceptors

In practice, semiconductors are **doped** with controlled levels of selected impurities. Donor impurities contribute electrons to the conduction band and produce n-type semiconductors. Acceptor impurities contribute holes to the valence band and produce p-type semiconductors.

The p-n *Junction*

A p-n junction (see Fig. 46-12) can serve as a diode rectifier. For forward-biasing (p positive with respect to n), the potential barrier is low, the junction is thin, and the forward current is large. For back-biasing, the potential barrier is high, the junction is thick, and the back current is small, usually negligibly so. The junction region itself, regardless of the applied potential difference, is called a **depletion layer.** It is virtually free of charge carriers and behaves like a somewhat leaky insulating slab.

The Light-Emitting Diode (LED)

A p-n junction can, under certain circumstances, convert the energy lost by a charge carrier crossing the barrier into visible light whose wavelength is

$$\lambda = \frac{c}{f} = \frac{hc}{E_g}, \qquad (46\text{-}9)$$

E_g being the energy gap width.

Transistors

A **transistor** is a three-terminal solid-state device in which the current flowing from the **drain** to the **source** is controlled by varying the potential of a **gate;** see Fig. 46-18. Figure 46-19 shows a MOSFET-type transistor, in which a small variation of the potential difference V_{GS} between the gate G and the source S has a major controlling effect on the current i_{DS} between the drain D and the source S.

28. A semiconductor contains equal numbers of donor and acceptor impurities. Do they cancel each other in their electrical effects? If so, what is the mechanism? If not, why not?

29. Germanium and silicon are similar semiconducting materials whose principal distinction is that the gap width E_g (see Fig. 46-8) is 0.67 eV for the former and 1.14 eV for the latter. If you wished to construct a p-n junction in which the back current is to be kept as small as possible, which material would you choose and why?

30. Consider two possible techniques for fabricating a p-n junction. (a) Prepare separately an n-type and a p-type sample and join them together, making sure that their abutting surfaces are plane and highly polished. (b) Prepare a single n-type sample and diffuse an excess acceptor impurity into it from one face, at high temperature. Which method is preferable and why?

31. In a p-n junction we have seen that electrons and holes may diffuse, in opposite directions, through the junction region. What is the eventual fate of each such particle as it diffuses into the material on the opposite side of the junction?

32. Does the diode rectifier whose characteristics are shown in Fig. 46-13 obey Ohm's law? What is your criterion for deciding?

33. We have seen that a simple intrinsic (undoped) semiconductor cannot be used as a light-emitting diode. Why not? Would a heavily doped n-type or p-type semiconductor work?

34. Explain how the MOSFET device of Fig. 46-19 works.

35. The acronym MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor. What is the significance of each of these terms as applied to the device shown in Fig. 46-19?

EXERCISES & PROBLEMS

SECTION 46-5 METALS: QUALITATIVE

1E. At what pressure, in atmospheres, would an ideal gas have a density of molecules equal to the density of the conduction electrons in copper ($= 8.43 \times 10^{28}$ m^{-3})? Assume $T = 300$ K.

2E. Gold is a monovalent metal with a molar mass of 197 g/mol and a density of 19.3 g/cm^3. Calculate the density of charge carriers.

3P. Calculate the number of particles per cubic meter for (a) the molecules of oxygen gas at 0°C and 1.0-atm pressure and (b) the conduction electrons in copper. (c) What is the ratio of these numbers? (d) What is the average distance between particles in each case? Assume that this distance is the edge of a cube whose volume is equal to the volume per particle. (See Sample Problem 28-3.)

4P. The density and molar mass of sodium are 971 kg/m^3 and 23 g/mol, respectively; the radius of the ion Na^+ is 98 pm. (a) What fraction of the volume of metallic sodium is available to its conduction electrons? (b) Carry out the same calculation for copper. Its density, molar mass, and ionic radius are, respectively, 8960 kg/m^3, 63.5 g/mol, and 135 pm. (c) For which of these two metals do you think the conduction electrons behave more like a free electron gas?

SECTION 46-6 METALS: QUANTITATIVE

5E. Use Eq. 46-6 to verify that the Fermi energy of copper is 7.0 eV. (Note, from Sample Problem 46-1, that the density of charge carriers in copper is 8.43×10^{28} m^{-3}.)

6E. (a) Show that Eq. 46-3 can be written as

$$n(E) = CE^{1/2},$$

where $C = 6.78 \times 10^{27}$ m$^{-3} \cdot$ eV$^{-3/2}$. (b) Use this relation to verify a calculation of Sample Problem 46-3, namely, that for $E = 5.00$ eV, $n(E) = 1.52 \times 10^{28}$ m$^{-3} \cdot$ eV^{-1}.

7E. Calculate the density $n(E)$ of conduction electron states in a metal for $E = 8.0$ eV and show that your result is consistent with the curve of Fig. 46-6*a*.

8E. What is the probability that a state 0.062 eV above the Fermi energy is occupied at (a) $T = 0$ K and (b) $T = 320$ K?

9E. The Fermi energy of copper is 7.0 eV. For copper at 1000 K, (a) find the energy at which the occupancy probability is 0.90. For this energy, evaluate (b) the density of states and (c) the density of occupied states.

10E. Show that Eq. 46-6 can be written as

$$E_F = An^{2/3},$$

where the constant A has the value 3.65×10^{-19} m$^2 \cdot$ eV.

11E. The density of gold is 19.3 g/cm^3. Each atom contributes one conduction electron. Calculate the Fermi energy of gold.

12E. Figure 46-7*c* shows the density of occupied states $n_o(E)$ of the conduction electrons in a metal at 1000 K. Calculate $n_o(E)$ for copper for the energies $E = 4.00$, 6.75, 7.00, 7.25, and 9.00 eV. The Fermi energy of copper is 7.00 eV.

13E. It can be shown that the conduction electrons in a metal behave like an ideal gas of the ordinary kind if the temperature is high enough. In particular, the temperature must be such that $kT \gg E_F$, the Fermi energy. What temperatures are required for copper ($E_F = 7.0$ eV) for this to be true? Study Fig. 46-7*c* in this connection and

note that we have $kT \ll E_F$ for the conditions of that figure. This is just the reverse of the requirement cited above. Note also that copper boils at 2595°C.

14E. The Fermi energy of silver is 5.5 eV. (a) At $T = 0$°C, what are the probabilities that states at the following energies are occupied: 4.4 eV, 5.4 eV, 5.5 eV, 5.6 eV, 6.4 eV? (b) At what temperature will the probability that a state at 5.6 eV is occupied be 0.16?

15E. The Fermi energy of aluminum is 11.6 eV; its density is 2.70 g/cm^3, and its molar mass is 27.0 g/mol (see Appendix D). From these data, determine the number of free electrons per atom.

16P. Show that the occupancy probabilities of two states whose energies are equally spaced above and below the Fermi energy add up to unity.

17P. Show that the probability p_h that a *hole* exists at a state of energy E is given by

$$p_h = \frac{1}{e^{-(E-E_F)/kT} + 1}.$$

(*Hint:* The existence of a hole means that the state is unoccupied; convince yourself that this implies that $p_h = 1 - p$.)

18P. Zinc is a bivalent metal. Calculate (a) the number of conduction electrons per cubic meter, (b) the Fermi energy E_F, (c) the Fermi speed v_F, and (d) the de Broglie wavelength corresponding to this speed. See Appendix D for needed data on zinc.

19P. Silver is a monovalent metal. Calculate (a) the number of conduction electrons per cubic meter, (b) the Fermi energy E_F, (c) the Fermi speed v_F, and (d) the de Broglie wavelength corresponding to this speed. Extract needed data from Appendix D.

20P. White dwarf stars represent a late stage in the evolution of stars like the sun. They become dense enough and hot enough that we can analyze their structure as a solid in which all Z electrons per atom are free. For a white dwarf with a mass equal to that of the sun and a radius equal to that of the Earth, calculate the Fermi energy of the electrons. Assume the atomic structure to be represented by iron atoms, and $T = 0$ K.

21P. A neutron star can be analyzed by techniques similar to those used for ordinary metals. In this case the neutrons (rather than electrons) obey the probability function, Eq. 46-7. Consider a neutron star of 2.0 solar masses with a radius of 10 km. Calculate the Fermi energy of the neutrons.

22P. Show that the density-of-states function given by Eq. 46-3 can be written in the form

$$n(E) = 1.5nE_F^{-3/2}E^{1/2}.$$

Explain how it can be that $n(E)$ is independent of the material when the Fermi energy E_F (= 7.0 eV for copper, 9.4 eV for zinc, and so on) appears explicitly in this expression.

23P. Estimate the number N of conduction electrons in a metal that have energies greater than the Fermi energy as follows. Strictly, N is given by

$$N = \int_{E_F}^{E_T} n(E)\,p(E)\,dE,$$

where E_T is the energy at the top of the band. By studying Fig. 46-7*c*, convince yourself that, to a good degree of approximation, this expression can be written as

$$N = \int_{E_F}^{E_F + 4kT} n(E_F)\left(\tfrac{1}{4}\right)\,dE.$$

By substituting the density of states function, evaluated at the Fermi energy, show that this yields for the fraction f of conduction electrons excited to energies greater than the Fermi energy,

$$f = \frac{N}{n} = \frac{3kT/2}{E_F}.$$

Why not evaluate the first integral above directly without resorting to an approximation?

24P. Use the result of Problem 23 to calculate the fraction of excited electrons in copper at temperatures of (a) absolute zero, (b) 300 K, and (c) 1000 K.

25P. At what temperature will the fraction of excited electrons in lithium equal 0.013? The Fermi energy of lithium is 4.7 eV. See Problem 23.

26P. Silver melts at 961°C. At the melting point, what fraction of the conduction electrons are in states with energies greater than the Fermi energy of 5.5 eV? See Problem 23.

27P. Show that, at the absolute zero of temperature, the average energy $\bar{E}$ of the conduction electrons in a metal is equal to $\frac{3}{5}E_F$, where E_F is the Fermi energy. (*Hint:* Note that, by definition of average, $\bar{E} = (1/n)\int En_0(E)\,dE$.)

28P. Use the result of Problem 27 to calculate the total translational kinetic energy of the conduction electrons in 1.0 cm^3 of copper at absolute zero.

29P. (a) Using the result of Problem 27, estimate how much energy would be released by the conduction electrons in a penny (assumed all copper; mass = 3.1 g) if we could suddenly turn off the Pauli exclusion principle. (b) For how long would this amount of energy light a 100-W lamp? Note that there is no known way to turn off the Pauli principle!

SECTION 46-8 DOPING

30E. The probability function of Section 46-6 can be applied to semiconductors as well as to metals. In semiconductors, E is the energy above the top of the valence band. The Fermi level for an intrinsic semiconductor is nearly midway between the top of the valence band and the bottom of the conduction band. For germanium these bands are separated by a gap of 0.67 eV. Calculate the probability that (a) a state at the bottom of the conduction band is occupied and (b) a state at the top of the valence band is unoccupied at 300 K.

31E. Pure silicon at room temperature has an electron density in the conduction band of approximately $1 \times 10^{16}\ m^{-3}$ and an equal density of holes in the valence band. Suppose that one of every 10^7 silicon atoms is replaced by a phosphorus atom. (a) Which type will this doped semiconductor be, *n* or *p*? (b) What charge carrier density will the phosphorus add? (See Appendix D for needed data on silicon.) (c) What is the ratio of the charge carrier density in the doped silicon to that in the pure silicon?

32E. What mass of phosphorus would be needed to dope a 1.0-g sample of silicon to the extent described in Sample Problem 46-5?

33P. Doping changes the Fermi energy of a semiconductor. Consider silicon, with a gap of 1.11 eV between the valence and conduction bands. At 300 K the Fermi level of the pure material is nearly at the midpoint of the gap. Suppose that it is doped with donor atoms, each of which has a state 0.15 eV below the bottom of the conduction band, and suppose further that doping raises the Fermi level to 0.11 eV below the bottom of that band. (a) For both the pure and doped silicon, calculate the probability that a state at the bottom of the conduction band is occupied. (b) Also calculate the probability that a donor state in the doped material is occupied. See Fig. 46-21.

34P. A silicon sample is doped with atoms having a donor state 0.11 eV below the bottom of the conduction band. (a) If each of these states is occupied with probability 5.00×10^{-5} at $T = 300$ K, where is the Fermi level relative to the top of the valence band? (b) What then is the probability that a state at the bottom of the conduction band is occupied? The energy gap in silicon is 1.11 eV.

35P. In a simplified model of an intrinsic semiconductor (no doping), the actual distribution in energy of states is replaced by one in which there are N_v states in the valence band, all of these states having the same energy E_v, and N_c states in the conduction band, all of these states having the same energy E_c. The number of electrons in the conduction band equals the number of holes in the valence band. (a) Show that this last condition implies that

$$\frac{N_c}{e^{(E_c - E_F)/kT} + 1} = \frac{N_v}{e^{-(E_v - E_F)/kT} + 1}.$$

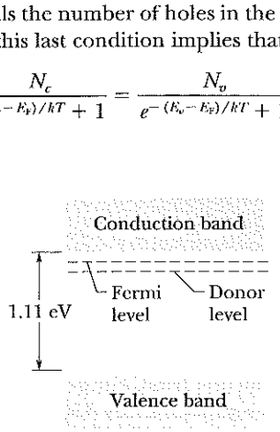


FIGURE 46-21 Problem 33.

(*Hint:* See Problem 17.) (b) If the Fermi level is in the gap between the two bands and is far from both bands compared to kT, then the exponentials dominate in the denominators. Under these conditions, show that

$$E_F = \tfrac{1}{2}(E_c + E_v) + \tfrac{1}{2}kT \ln(N_v/N_c),$$

and therefore that, if $N_v \approx N_c$, the Fermi level is close to the center of the gap.

SECTION 46-9 THE *p-n* JUNCTION

36E. When a photon enters the depletion region of a *p-n* junction, electron–hole pairs can be created as electrons absorb part of the photon's energy and are excited from the valence band to the conduction band. These junctions are thus often used as detectors for photons, especially for x rays and nuclear gamma rays. When a 662-keV gamma-ray photon is totally absorbed by a semiconductor with an energy gap of 1.1 eV, on the average how many electron–hole pairs are created?

37P. For an ideal *p-n* junction diode, with a sharp boundary between the two semiconducting materials, the current i is related to the potential difference V across the diode by

$$i = i_0(e^{eV/kT} - 1),$$

where i_0, which depends on the materials but not on the current or potential difference, is called the *reverse saturation current.* V is positive if the junction is forward-biased and negative if it is back-biased. (a) Verify that this expression predicts the behavior expected of a diode by sketching i as a function of V over the range $-0.12\ V < V < +0.12\ V$. Take $T = 300$ K and $i_0 = 5.0$ nA. (b) For the same temperature, calculate the ratio of the current for a 0.50-V forward-bias to the current for a 0.50-V back-bias.

SECTION 46-11 THE LIGHT-EMITTING DIODE (LED)

38E. (a) Calculate the maximum wavelength that will produce photoconduction in diamond, which has a band gap of 7.0 eV. (b) In what part of the electromagnetic spectrum does this wavelength lie?

39E. In a particular crystal, the highest occupied band of states is full. The crystal is transparent to light of wavelengths longer than 295 nm but opaque at shorter wavelengths. Calculate, in electron-volts, the gap between the highest occupied band and the next (empty) band.

40E. The KCl crystal has a band gap of 7.6 eV above the topmost occupied band, which is full. Is this crystal opaque or transparent to light of wavelength 140 nm?

41P. Fill in the seven-segment display shown in Fig. 46-16*a* to show how all 10 numbers may be generated. (b) If the numbers are displayed randomly, in what fraction of the displays will each of the seven segments be used?

THE LIQUID CRYSTAL STATE OF MATERIALS, OR, HOW DOES THE LIQUID CRYSTAL DISPLAY (LCD) IN MY WRISTWATCH WORK?

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ESSAY 17



Patricia Elizabeth Cladis was born in Shanghai, China, and grew up in Vancouver, British Columbia. She received her Ph.D. in physics from the University of Rochester with a thesis on the dc superconducting transformer. Before joining AT&T Bell Laboratories, she did postdoctoral research at the University of Paris, Orsay, France, where she first learned about liquid crystals and discovered "escape into the third dimension" and point defects in nematics. At Bell Labs she discovered the "reentrant nematic" phase. Currently she uses liquid crystals to study general problems in nonlinear physics. She has published nearly 100 scientific papers and is on the editorial board of the journal Liquid Crystals.

More than 20 years ago, while at the Liquid Crystal Institute at Kent State University, Kent, Ohio, James Fergason realized that the large optical response of liquid crystals to small voltages could be used as a key component in electronic devices. A new technology was born as complex organic materials, the building blocks of nature, made their debut in the electronics industry as flat panel displays known as LCDs—liquid crystal displays. The first applications for LCDs were as low-information-density, alphanumeric displays for electronic products such as wristwatches, calculators, and "smart boxes" like the surveillance system shown in Fig. 1.

With increased understanding of their optoelectronic properties, LCDs moved toward higher-information-density applications involving pictures, as well as numbers and letters, required, for example, by lap-top PCs (Fig. 2) and color TVs. Liquid crystal color displays are awesome and need much less space than conventional displays. The LCD display package is flat and lightweight, and consumes little electrical power even when backlighting is added, as in the display shown in Fig. 2.

The purpose of this essay is to outline the physics involved in the operation of one picture element, or *pixel*, in the simplest LCD, the twisted nematic display. While the basic phys-

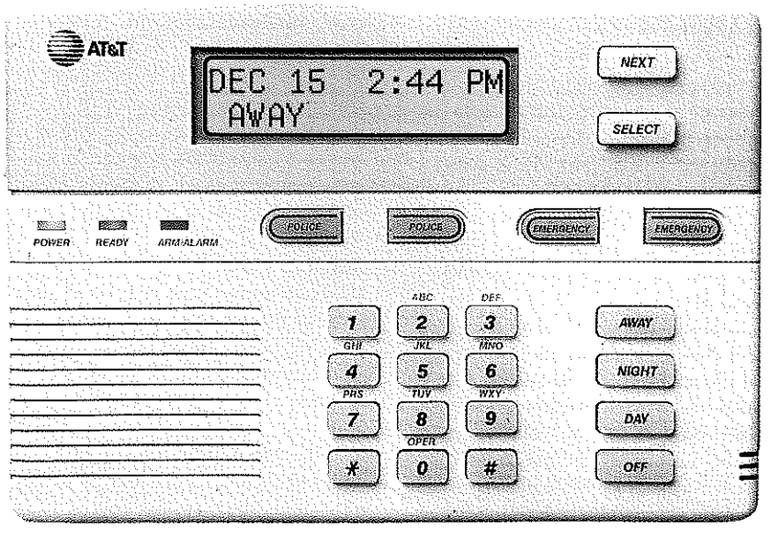


FIGURE 1 Alphanumeric LCDs easily transfer information between fast "smart" electronic services and human customers.

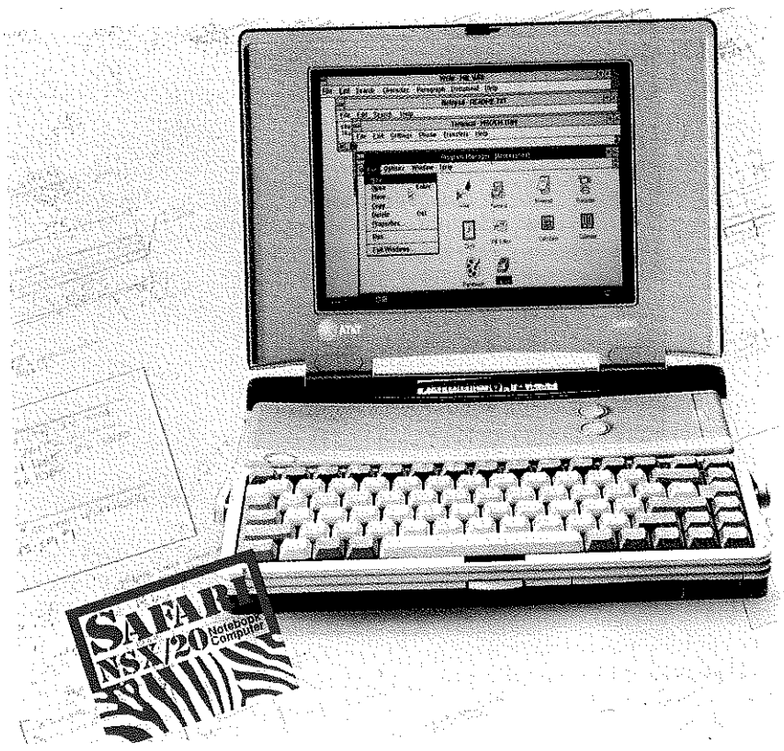


FIGURE 2 High-information-density LCDs showing pictures, as well as numbers and letters, are used in lap-top PCs.

ics in the operation of one pixel is, roughly speaking, the same as in high-information-density displays, the passage from low-density to high-density displays required considerable R&D effort involving chemists, optoelectronic engineers, physicists,* materials scientists, and mathematicians, as well as the innovation of sophisticated manufacturing processes. Because of the expanding role computers play in our daily life, it is expected that before the turn of the century, LCDs will dominate the high-information-density display market.

Molecules That Align

In a crystal, molecules and atoms are organized in a three-dimensional structure where each atom or molecule does not move far from its position in the structure. Liquid phases have no structural order, so, while keeping close to each other, molecules or atoms move freely in the liquid state. Molecules are made up of many atoms and can exhibit a new state of matter called *liquid crystal* that is not as ordered as a solid crystal but also not as disordered as the usual liquid state, referred to as the *isotropic liquid* state.

Molecular shape plays an important role in determining liquid crystal phases. A molecule can be spherical, rodlike or cigar-shaped, disklike, bowl-shaped, or some other complex shape. Liquid crystal phases form when rod-shaped molecules spontaneously align while remaining liquid. In the liquid crystal state called *nematic*, which is used in the displays shown in Figs. 1 and 2, a degree of alignment is maintained despite rapid, random thermal motion of individual molecules characteristic of the liquid state. The property of certain molecules to align is called *long-range orientational order*. It is the property that characterizes liquid crystals.

*Indeed, P. G. de Gennes was awarded the 1991 Nobel prize in physics for his work on liquid crystals and polymers.

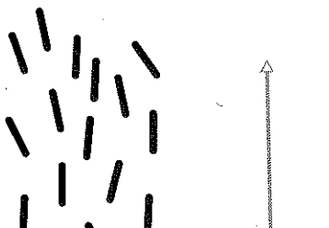


FIGURE 3 Structure of the nematic liquid crystal phase: rodlike molecules align. The alignment is not perfect because molecules in the liquid state are in rapid, random thermal motion.

Molecules forming the nematic liquid crystal used in displays are 20 Å long and about 5 Å in diameter. The nematic state is a result of many molecules aligning cooperatively in the same direction. In discussing liquid crystal phases, one thinks in terms of many molecules, not one. Instead of referring to a single molecule, we talk about the collection of molecules in the nematic state and refer to the direction of alignment as the *director* or, in short-hand form, **n**, where **n** is a unit vector oriented in the direction chosen by many molecules for alignment. It is illustrated by lines, such as the ones shown in Fig. 3. The direction of alignment, **n**, defines an *optic axis* for materials in the nematic state.

Important physical principles that determine how LCDs work are as follows:

1. The orientation of **n** can be determined by small forces such as weak electric fields or surface forces.
2. When two different forces compete for the orientation of **n**, the response of **n** can be tuned by varying one of the forces.

tic

Alignment direction **n**

Structure of the nematic phase: rodlike molecules gnment is not perfect be- ules in the liquid state are dom thermal motion.

m thermal motion of in- ecules characteristic of ate. The property of cer- es to align is called *long- ional order.* It is the prop- racterizes liquid crystals. es forming the nematic used in displays are 20 Å out 5 Å in diameter. The e is a result of many mol- ng cooperatively in the on. In discussing liquid s, one thinks in terms of ules, not one. Instead of a single molecule, we talk llection of molecules in state and refer to the di- ignment as the *director* or, d form, **n**, where **n** is a riented in the direction any molecules for align- lustrated by lines, such as wn in Fig. 3. The direc- ment, **n**, defines an *optic* erials in the nematic state. ant physical principles ine how LCDs work are as

ntation of **n** can be deter- nall forces such as weak ds or surface forces.

o different forces com- orientation of **n**, the re- can be tuned by varying orces.

3. The behavior of polarized light traveling parallel to **n** is different from that of polarized light traveling perpendicular to **n**.

Determining the Orientation of n

Surface forces provide a useful way to select **n**. For example, when microscope slides are buffed in a single direction many times on a piece of white filter paper, a nematic liquid crystal in contact with the buffed surface orients with **n** following the buffing direction. Conventional wisdom is not that this is magic but rather that oil from the fingers is transferred to the glass surfaces. The buffing process produces microscopic grooves in the oil, providing an easy direction for the alignment of **n**.

To make a liquid crystal sample with a uniform **n**, the nematic liquid crystal is sandwiched between two buffed glass plates with the buffed surfaces facing the liquid and the directions of buffing on the two plates parallel to each other. In such a sample, **n** is parallel to the glass surfaces and parallel to the buffing direction throughout the whole sample.

Another way to orient **n** is with an *electric field* **E**. In some materials, **n** aligns parallel to **E** while in others, **n** aligns perpendicular to **E**. Materials for which **n** aligns parallel to **E** are called *positive materials.* The vector **n** orients in both ac and dc fields as there is no energy difference between the state **n** parallel to **E** and the state **n** antiparallel to **E**. Typically, LCDs use ac fields and positive materials.

When surface forces at electrodes pin **n** perpendicular to **E**, then a large-enough voltage has to be applied to a positive material before **n** responds: a *threshold voltage* $V_c = E_c/d$ (d is the distance between the electrodes) has to be applied for **E** to successfully compete with surface forces in the reorientation of **n**. For $V > V_c$, the component of **n** parallel to **E** is determined by the relative strengths of the applied electric field and the pinning force. A value of $V_c \approx 2$ V is a typical threshold voltage for nematics. As this is small by industry standards, relatively cheap electronics are needed to drive LCDs.

The Twisted Nematic

This is the configuration widely used in LCDs. A positive nematic liquid crystal is sandwiched between buffed glass plates that have transparent electrodes evaporated onto the buffed sides. The two plates are oriented with their respective buffing directions nearly perpendicular to each other. The director **n** twists nearly 90° smoothly between the two plates. A typical distance between electrodes is 6 μm. The polarization direction of incoming polarized light follows the gentle twist in **n**. When $V < V_c$, a pixel in the twisted state looks bright when viewed between crossed polarizers. The upper part of Fig. 4 shows this.

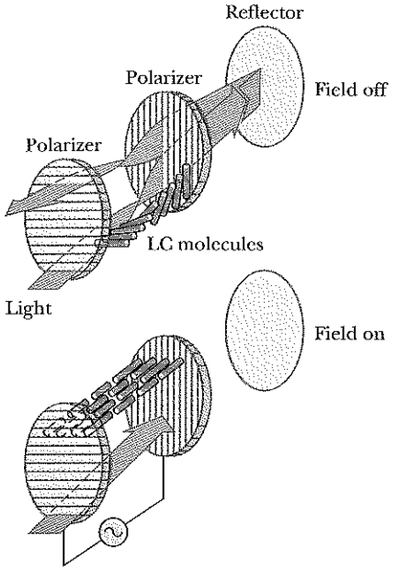


FIGURE 4 The simplest liquid crystal display uses the twisted nematic configuration. In the fully off state, shown in the upper part of the figure, the director twists nearly 90° between electrodes. The twisted structure is destroyed when **n** aligns parallel to **E**. In LCDs, the twisted nematic is sandwiched between crossed polarizers. A mirror reflects light back through the twisted structure to form the white background of the display. To display a character, an applied electric field destroys the twisted structure and no light reaches the mirror to be reflected back. The fully on state is shown in the bottom part of the figure.

When $V \gg V_c$, the optic axis is parallel to the electric field, destroying the twisted structure. The polarization direction of incoming polarized light does not rotate as the light traverses the liquid crystal and is extinguished by the second polarizer. When $V \gg V_c$, the pixel is as dark as allowed by the crossed polarizers. This is shown in the lower part of Fig. 4: the state used to show characters on a monochrome display. Intermediate shades of grey are observed for intermediate values of V. When the electric field is turned off, the orientation of **n** is again determined by surface forces and **n** relaxes back to the initially twisted configuration. The relaxation time depends on the distance between the electrodes and on an orientational diffusion constant.

Liquid crystal displays require little power because ambient light is used. Even with additional power consumption from backlighting used in higher-information-density displays, such as Fig. 2, the total power to run LCDs is smaller than conventional displays. Another LCD feature is that it works in white light, and so color displays can be made by adding color filters to the glass substrates. Indeed, liquid crystals make stunning color displays. Typical LCD switching times are about 10 ms. For some applications this is too slow. A major part of LCD research is to find ways to reduce this time.

The Effect of Temperature

Liquid crystals are useful for studying the general problem of how order in matter is created and destroyed. Indeed, some scientists are using nematic liquid crystals to model the birth of the universe immediately after the Big Bang! Still others are using liquid crystal materials to learn about nonlinear, nonequilibrium dynamic processes and transitions to chaos.

When rodlike molecules align, they are more densely packed than when randomly oriented. For example, matches arranged in a box

are more closely packed than when thrown at random onto a table. At some low range of temperatures, rod-shaped molecules prefer to pack in the denser aligned state. As temperature is increased, the system becomes less dense and more energetic. The chance that all the molecules choose the same alignment is reduced. At a special temperature, called the *transition temperature,* there is a sudden change to another liquid state—to another phase—without long-range orientational order. A phase transition takes place from the nematic liquid crystal phase to the isotropic liquid.

This transition is easily observed when a nematic liquid crystal* is heated. In samples a few millimeters thick, the nematic state is translucent, like frosted glass, because of fluctuations in the alignment caused by thermal motion of the molecules. These fluctuations cause *local variations* in the index of refraction that scatter light. When heated into the isotropic liquid state, a thick sample becomes transparent because the molecules are now *uniformly* disordered.

Thus the LCD operates in a certain temperature range. If the temperature is too high, the material transforms to the isotropic liquid, losing orientational order, and the display loses contrast. If the temperature is too low, the material transforms to a more ordered liquid crystal state or, perhaps, even the crystalline state, and its orientation cannot be easily changed. Although materials are known that have a nematic state from $-50°C$ to $+400°C$, the nematic temperature range in any one compound is typically between 1 K and 20–30 K. To obtain the wide temperature range needed for applications, several different compounds are mixed together.

*Liquid crystals may be purchased from many chemical companies, such as EM Industries, 5 Skyline Drive, Hawthorne, NY 10532 and Roche Vitamins and Fine Chemicals, 340 Kingsland Street, Nutley, NJ 07110.

In a flat panel display such as used in today's lap-top computers, picture information reaches the screen through rows of electrodes on one substrate and columns of electrodes on the other. The intersection of the electrodes forms a grid of pixels. In a simple matrix drive system, electric signals are applied to the row and column electrodes with the proper timing to select the target pixel. Because a pixel responds to the rms voltage on a line, an *iron law* prevails that limits the number of lines that can be addressed. In the case of the twisted nematic, this is about 200 lines. When the nematic is twisted even more (between 180° and 270°, now called a *super-twisted* nematic, STN, display), the electro-optic response becomes steeper, making 768 lines available. But to make an STN display requires a liquid crystal that spontaneously twists: a chiral liquid crystal.

Chiral Liquid Crystals

In the twisted nematic display, the twist in the optic axis is determined by the surface treatment of the two glass plates. If a twist of more than 90° is applied, say, 120°, it costs less energy for **n** in nematic liquid crystals to satisfy this boundary condition by twisting only 60°. A nematic does not spontaneously twist and will always minimize the amount of twist applied by boundary conditions. When the display used is an STN display, where the twist is between 180° and 270°, the display contrast is excellent for a much wider viewing angle.

As **n** in a nematic will not twist through angles larger than 90°, a liquid crystal phase that spontaneously twists is used in STN displays. These materials are known as cholesteric liquid crystals because they were first derived from natural products like butter and cheese. Unlike the nematic display, where the director twists nearly 90° over a distance determined by the electrode separation, in cholesteric liquid crystals, **n** spontaneously twists 360° in a distance called the pitch that is a characteristic of the material. Cholesteric liquid crystals are one example of *chiral* liquid crystal phases. The nematic is the special case of a cholesteric where the pitch is infinite.

The two ways of twisting are identified as left-handed and right-handed. In the schematic shown in Fig. 4, the director twists in a left-handed manner. To see this, extend the thumb of the left hand in the direction of twist, that is, perpendicular to the glass plates with thumb tip toward the second plate. On going through the liquid from the plate called "one" to the plate called "two," fingers of the left hand naturally curl to maintain alignment with the director. Does it matter which plate is called "one"? Fingers of the right hand curl in the opposite sense and follow a right-handed twist. It costs the same energy for a nematic liquid crystal to twist in a left-handed sense as in a right-handed sense. It costs no energy for a cholesteric liquid crystal to twist in one sense and a great deal to twist in the other.

Figure 5 shows half a pitch of the left-handed cholesteric structure. To make clearer the three-dimensional property of this helical structure, cylinders, rather than the simple lines used in Fig. 3, are used to depict **n** in Fig. 5. As cholesterics are also three-dimensional liquids, the cylinders are not on perfect rows or columns. A left hand is also shown in Fig. 5 to illustrate how handedness is determined. In the same way that the mirror image of a right hand is a left hand, the mirror image of a right-handed twisted structure is a left-handed one.

Cholesteric liquid crystals have an interesting way of interacting with light. They are *optically active.* While it is nontrivial to synthesize optically active materials in the laboratory, nature does it all the time. Indeed, in a Dorothy Sayers detective story, a deadly case of murder by mushroom poisoning was solved when it was discovered that the mushroom poison ingested by the victim was not opti-

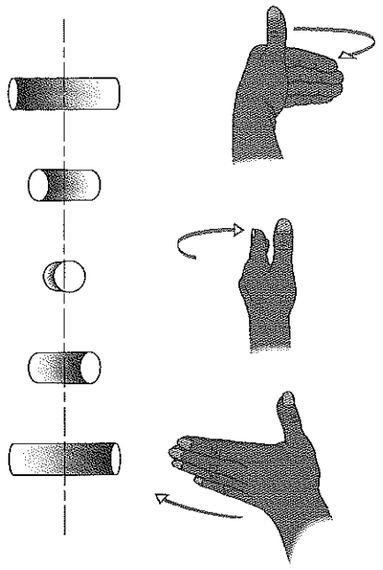


FIGURE 5 The structure of cholesteric liquid crystals. To capture the three-dimensional property of its helical structure, cylinders rather than lines are used to show the director orientation in the structure. The director **n** is perpendicular to the twist direction and rotates 360° uniformly in a fixed distance called the pitch. In the figure, half a pitch is shown. There are no layers in the cholesteric phase. The rotation sense of **n** is the same as the direction the fingers of one hand curl, shown by the arrows, when the thumb is extended in the direction **n** twists.

cally active! It had been synthesized in a university chemistry laboratory—not by the murderer, it turned out.

Unpolarized white light can be thought of as composed of both left and right *circular* polarizations rotating at all the frequencies corresponding to all the wavelengths constituting white light. A cholesteric liquid crystal with a pitch comparable to the wavelength of light transmits all the incident light *except* circularly polarized light with the *opposite* hand to its twisted structure and of a particular wavelength corresponding to its pitch and the angle of incidence of the light. When light travels in the direction of twist, the wavelength equal to the pitch is rejected. Light traveling obliquely to the twist direction "sees" a shorter pitch, and so a shorter wavelength is scattered back. Viewed in ordinary diffuse daylight, the scattering of different wavelengths in different directions produces a striking display of vivid colors "recalling the appearance of a peacock's feathers."

In some materials the pitch depends sensitively on temperature. These materials are used as thermometers to translate small temperature differences into different colors. In some countries, newborns in a nursery have a small adhesive patch of encapsulated temperature-sensitive cholesteric on their foreheads. A single nurse can identify at a glance when one of them has a fever. A temperature map of complicated surfaces —for example, parts of the human body—is obtained by spraying a thin cholesteric liquid crystal layer onto the surface in question. In the medical field, temperature-sensitive cholesterics are an inexpensive, noninvasive way to detect at an early stage tumors, cancerous growth, and, in some Third World countries, leprosy, as the temperature of these tissues is usually different from the temperature of healthy tissue.

Discussion Topics

1. How would the two states of the LCD in Fig. 4 look if the polarizers were parallel to each other?

2. An idealized LCD is shown in Fig. 4 with the director exactly parallel to the glass plates. This is not practical for a display because there are two ways for **n** to turn toward **E**. Why is this bad for a display? Draw the picture with **n** rotated by a small angle θ toward **E** in one region and $-\theta$ in another. How do the two regions connect? This problem is overcome by treating the glass substrates so that **n** is inclined a little to them. Why does this work? How do you think a small angle affects the sharpness of the optical response above threshold?

3. Harder question. A little bit of cholesteric material is added to most twisted nematic displays to ensure that only one hand is preferred in the twisted structure. Why is this a good idea? Draw a picture of a left-handed twist in one part of the sample and a right-handed one in the other. How do they connect up?

4. While V_c does not depend on the distance between electrodes, how does the time to turn off a pixel depend on this distance? *Hint*: The dimensions of a diffusion constant, D_0, are square centimeters per second and the distance between electrodes, d, is in centimeters. Put D_0 and d together to get a quantity that has units of time.

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